

# The Solar System at Ultraviolet Wavelengths

Amanda R. Hendrix

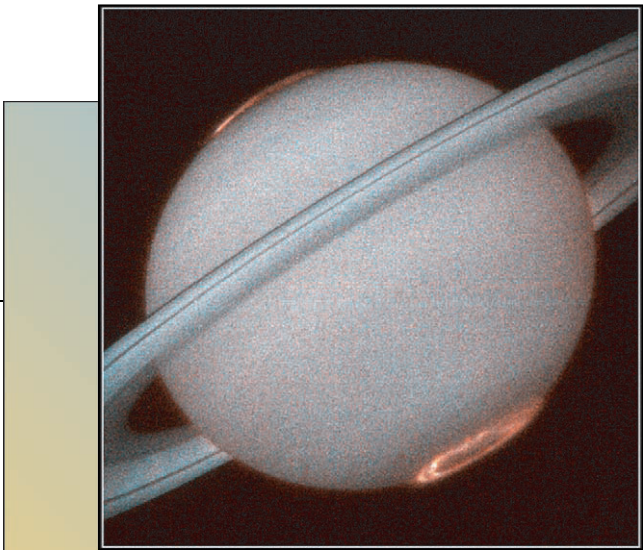
Jet Propulsion Laboratory  
California Institute of Technology  
Pasadena California

Robert M. Nelson

Jet Propulsion Laboratory  
California Institute of Technology  
Pasadena California

Deborah L. Domingue

Applied Physics Laboratory  
Johns Hopkins University  
Laurel, Maryland



## CHAPTER 36

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Ultraviolet imaging and spectroscopy are powerful tools for probing planetary atmospheres and surfaces. In this chapter, we review the significant contributions to our understanding of the solar system that have been made by ultraviolet observing methods. We cover results from the near-ultraviolet (NUV, 2000–3500 Å), far-ultraviolet (FUV, 1000–2000 Å), and extreme-ultraviolet (EUV, 500–1000 Å) wavelength ranges. These are shorter than visible and near-infrared wavelengths and involve photons of increasingly higher energy. Ultraviolet observations therefore provide unique insight into planetary processes involving more energetic processes that cannot be studied using photons of longer wavelength. Many of the solar system observations in the ultraviolet have been performed by Earth-orbiting telescopes, such as the *International Ultraviolet Explorer (IUE)* and the *Hubble Space Telescope (HST)*. We also review results from ultraviolet instruments on *Galileo*, *Cassini*, *Voyager*, and other spacecraft. Each planet in the solar system except Mercury has been observed in the ultraviolet by Earth-orbiting telescopes. Many of the larger planetary satellites, selected asteroids, and comets have also been observed. This data set has provided important information regarding the atmospheres and surfaces of solar system objects and the processes shaping their compositions.

### 1. A Brief History of Ultraviolet Astronomy

The ultraviolet spectral region is important to the entire community of astronomers, from those who study nearby objects such as Earth’s Moon to those who study objects at the edge of the observable universe. From the perspective of a planetary astronomer, the spectral information is important for determining the composition of, and understanding the physical processes that are occurring on, the surfaces and atmospheres of solar system objects.

Prior to the dawn of the space age, spectrophotometry of solar system objects at wavelengths shorter than ~3000 Å had long been desired in order to complement observations made by ground-based telescopes at longer wavelengths. However, the presence in Earth’s atmosphere of ozone, a strong absorber of ultraviolet light between 2000 and 3000 Å, and molecular oxygen (O<sub>2</sub>), which is the dominant ultraviolet absorber below 2000 Å, prevented astronomers of the 1950s and earlier from observing the universe in this important spectral region.

The ultraviolet wavelengths of astronomical sources became observable midway through the last century when instruments could be deployed above Earth’s atmosphere. A rocket or spacecraft provides a platform from which astronomical observations can be made where the light being

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collected has not been subjected to absorption from Earth's atmospheric gases. Thus, the space revolution dramatically enhanced the ability of astronomers to access the full spectrum of electromagnetic radiation emitted by celestial objects.

In the 1950s, a series of rocket-flown instruments began slowly to reveal the secrets of the ultraviolet universe. The first photometers and spectrometers were flown on unstabilized Aerobee rockets. They remained above the ozone layer for several tens of minutes while they scanned the sky at ultraviolet wavelengths. By the early 1960s, spectrometers on three-axis-stabilized platforms launched by rockets on suborbital trajectories were able to undertake observations with sufficient resolution such that individual spectral lines could be resolved in the target bodies.

Shortly thereafter, the military spacecraft designated 1964-83C carried an ultraviolet spectrometer into Earth orbit. This was followed closely by NASA's launch of the first Orbiting Astronomical Observatory (OAO) satellite in 1966. These space platforms permitted long-duration observations compared to what was possible from a rocket launch on a suborbital trajectory. By 1972, the third spacecraft of the OAO series was launched. It was designated the *Copernicus* spacecraft and was an outstanding success.

In Europe, a parallel pattern of development for exploring the ultraviolet sky was under way using sounding rockets followed by orbiting spacecraft. In 1972, the European Space Research Organization launched an Earth-orbiting spacecraft (TD-1A) dedicated to ultraviolet stellar astronomy. Such developments set the stage in the 1970s for a joint U.S.–European collaboration, the *International Ultraviolet Explorer* satellite.

The *IUE* spacecraft was launched in 1978 into a geosynchronous orbit over the Atlantic Ocean. From there it could be controlled from ground stations in Greenbelt Maryland in the United States or in Villefranca, Spain, by engineers from NASA or ESA. It functioned continuously from launch until it was terminated in 1996 and its capabilities taken over by instruments on the *Hubble Space Telescope*. *IUE* spectra were recorded in two wavelength ranges of 1150–1950 Å and 1900–3200 Å, at either high or low spectral resolution. *IUE* had no imaging capability, though spatial discrimination was possible within the largest (10 × 20 arcsec oval) spectrograph entrance aperture.

Additional Earth-orbiting satellites with ultraviolet observing capabilities were launched in the early 1990s. These include NASA's *Extreme Ultraviolet Explorer* satellite (*EUVE*) and the joint U.S.–European *Hubble Space Telescope*. *HST* is in a low-Earth orbit, allowing upgrades to the facilities by astronauts. However, the low orbit reduced the observational duty cycle to 50–60% of that of *IUE* in high orbit. UV spectroscopy with *HST* has been performed with the Goddard High-Resolution Spectrograph (GHRS), the Faint-Object Spectrograph (FOS), the Space Telescope Imaging Spectrograph (STIS), and the Advanced Camera

for Surveys (ACS). In 1990 and 1995, the *Hopkins Ultraviolet Telescope* (*HUT*) was flown aboard the U.S. space shuttle as part of the Astro Observatory. The *Far-Ultraviolet Spectroscopic Explorer* (*FUSE*) was launched in 1999 and has spectroscopic capabilities in the 900–1200 Å wavelength range.

Many interplanetary spacecraft missions have included ultraviolet instruments in their payloads. *Pioneers 10* and *11*, which were launched in 1970 and 1973, respectively, included ultraviolet photometers among their scientific instruments. [See PLANETARY EXPLORATION MISSIONS.] These two spacecraft were the first to safely pass through the Asteroid Belt and fly by Jupiter and Saturn. *Mariner 6* and *7*, Mars flyby missions launched in 1969, and *Mariner 9*, the first spacecraft to orbit Mars, launched in 1971, all carried ultraviolet spectrometers. *Mariner 10*, which flew by Mercury three times in 1974 and 1975, carried two extreme ultraviolet spectrometers (an **airglow** spectrometer and an occultation spectrometer) to measure the planet's exospheric composition. *Mariner 10* also made measurements of Earth's Moon after launch. *Pioneer Venus*, which was launched in 1978, was the first U.S. mission dedicated to the exploration of the planet Venus. It included an ultraviolet spectrometer among its instrument package. Soviet spacecraft missions *Vega 1* and *Vega 2*, launched in 1985, dropped two descent probes into Venus' atmosphere, which included a French–Russian ultraviolet spectroscopy experiment. The *Voyager* project sent two spacecraft that included ultraviolet spectrometers within their instrument payloads, both launched in 1977, to the outer solar system. *Voyager 2* was the first spacecraft to fly by all four of the jovian planets (Jupiter, Saturn, Uranus, and Neptune). In 1989, the *Galileo* spacecraft was launched. This spacecraft was the first dedicated mission to the Jupiter system, and it included within its scientific instrument payload two ultraviolet spectrometers, the EUV (extreme ultraviolet spectrometer, which operated between 500 and 1400 Å) and the UVS (the ultraviolet spectrometer that covered the 1150–4300 Å wavelength range). En route to Jupiter, the *Galileo* spacecraft collected ultraviolet spectra as it flew by Venus, the Moon, and the asteroids Gaspra and Ida. The *Cassini* mission, launched in 1997, includes the Ultraviolet Imaging Spectrograph (UVIS) and arrived at Saturn for a 4 year tour in June 2004. The *Nozomi* spacecraft, launched in 1998, carried two UV instruments; measurements were made of the Moon en route to Mars. (*Nozomi* unfortunately failed to enter Mars orbit.) *Mars Express*, in orbit since December 2003, has an ultraviolet instrument called SPICAM as part of its payload. In 2004, the *MERCURY Surface, Space ENVIRONMENT, GEOchemistry, and Ranging* (*MESSENGER*) mission launched on its way to Mercury carrying an ultraviolet-visible (1150 to 6000 Å) spectrometer. Also in 2004, Rosetta with its ALICE ultraviolet instrument (covering 700–2050 Å) was launched en route to comet 67P/Churyumov–Gerasimenko, set to enter orbit

in 2014. The *New Horizons* mission to Pluto, launched in January 2006, carries the next-generation version ALICE instrument, covering 520–1870 Å. *Venus Express* has been in orbit at Venus since April 2006 and employs the SPI-CAV/SOIR ultraviolet and infrared instrument for atmospheric studies.

## 2. Nature of Solar System Astronomical Observations

Most astronomers observe objects that have their own intrinsic energy source, such as stars and galaxies. However, the majority of the observations undertaken by planetary astronomers are of targets that do not emit their own radiation but are observable principally because they reflect the sunlight that falls on them or emit energy as a result of various physical processes. The measured spectrum of a body can thus reveal significant information on the composition of, and processes occurring within, planetary surfaces and atmospheres. The measured spectrum includes absorption features that can determine or constrain the composition of a surface or atmosphere, or emission features that suggest excitation processes in a gas or thermal emission from solids. The measured spectrum often displays solar features (either emission features or spectral continuum). To study the spectrum of the body itself, the solar spectrum is divided out, resulting in what is known as the spectral reflectance. The variation of the reflectance or **geometric albedo** (the reflectance at zero phase angle) as a function of wavelength is used to measure the strength of absorption features, from which the abundance of spectrally active species can be estimated.

Measuring reflected light at ultraviolet wavelengths can pose some interesting problems for instrument designers. First, instrument spectral sensitivity becomes weaker with decreasing wavelength and so does the Sun's energy output. The energy output of the Sun changes by a factor of  $10^3$  between the EUV and NUV spectral ranges, which until recently exceeded the dynamic range of UV detectors.

Second, in order to obtain the spectral reflectance of a body, a solar spectrum must be measured, which is no easy task in the ultraviolet range. Furthermore, below 1800 Å, the spectrum of the Sun is variable. Therefore, a simultaneous spectrum of the Sun (or the reflection spectrum from an object whose spectrum is well understood) must be gathered at the same time that any ultraviolet observations are undertaken.

Lastly, particularly when performing measurements from an Earth-orbiting observatory such as *IUE* or *HST*, solar system objects change positions against the background of stars during the course of an individual observation. In most cases, special tracking rates must be calculated prior to each observing run in order to know the change of the

position of the target with time. Inaccurately calculated tracking rates can cause the observed target to drift from the instrument's field of view, thus adding noise and uncertainty to a measurement.

## 3. Observations of Planetary Atmospheres

With the exception of the innermost planet Mercury (which possesses a surface-bounded exosphere, similar to that observed on the Moon), all the planets in the solar system (and a few planetary satellites) are surrounded by detectable atmospheres. All the planets with atmospheres absorb ultraviolet light, and as a result ultraviolet observations provide information on the composition of, and processes that are occurring in, the object's atmosphere.

In general, the atmospheres of the terrestrial planets (Mercury, Venus, Earth, and Mars) are considered secondary atmospheres because they evolved after the primordial atmospheres were lost. However, the atmospheres of the four jovian planets (Jupiter, Saturn, Uranus, and Neptune), because of their strong gravitational attraction and comparatively low temperatures, retained the primordial elements, particularly hydrogen and helium. From ground-based observations, methane ( $\text{CH}_4$ ) and ammonia ( $\text{NH}_3$ ) were identified in the atmospheres of the giant planets and therefore atmospheric processes were suspected of producing a host of daughter products that can be detected at ultraviolet wavelengths. [See ATMOSPHERES OF THE GIANT PLANETS.]

Sunlight entering a planetary atmosphere can experience or initiate a wide variety of processes that contribute to the total energy emitted by the object and observed by an astronomical facility. The objects described previously all possess atmospheres that contribute significantly to their spectral behavior. Astronomical observations of such bodies search for and measure the depths of absorption bands in the spectrum or emission bands due to atmospheric interactions with both solar photons and energetic particles that originate from the solar wind or the planet's magnetosphere. These bands are unique to specific gases; thus, it is possible to identify or eliminate particular gases as candidate materials in the atmospheres of these objects. The interpretation of an ultraviolet spectrum can be an arduous task, given that the bands and lines observed in the spectrum may arise from a combination of processes. These include:

1. Single and multiple scattering of photons by aerosols such as haze and dust (**Mie scattering**) and gas (**Rayleigh scattering/Raman scattering**) in the planetary atmosphere.
2. Absorption of the incident ultraviolet solar light by atmospheric species.



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3. Stimulation of an atmospheric gas by incident sunlight and emission by **fluorescence**, chemiluminescence, or resonant scattering.
4. Photoionization and photodissociation reactions that produce a reaction product in an excited state.
5. Excitation of gas by precipitation of magnetospheric particles.

Each of these processes is associated with a well-understood physical mechanism, the details of which are beyond the scope of this chapter. The reader is referred to the Bibliography and other chapters in this volume.

Two significant methods of studying atmospheres at UV wavelengths are stellar occultations (observing a star as a body passes in front of the star and measuring the stellar flux as it is diminished) and reflection/airglow measurements (measurements of the backscatter of the solar continuum either by Rayleigh–Raman atomic/molecular scattering or by Mie scattering from atmospheric aerosols). The atmospheric species and density can be constrained by studying the occulted stellar spectrum as it passes through the atmosphere. Limited-wavelength facilities can identify some but not all of the constituents present and processes ongoing in a planetary atmosphere. The ultraviolet data from Earth-orbiting satellites have been used in combination with ground-based observations at other wavelengths and with observations by other spacecraft (including flyby missions) to develop an understanding of the atmospheres of planetary objects. The following discussion summarizes the results of those bodies in the solar system that possess atmospheres.

### 3.1 Mercury and the Moon

Both Mercury and the Moon have very tenuous atmospheres that are often referred to as surface-bounded exospheres. The atoms in these atmospheres do not collide with each other; rather, they bounce from place to place on the surface. The *Mariner 10* ultraviolet airglow experiment detected hydrogen, helium, and oxygen atoms as constituents in Mercury's exosphere. No molecules were detected. The pressure of Mercury's atmosphere was determined to be about  $10^{-12}$  bar (compared to the 1 bar atmospheric pressure at sea level on Earth). Ground-based telescopic observations in the visible have identified resonant scattering emission features attributed to sodium, potassium, and calcium as well. Observations of Mercury's exospheric sodium demonstrate that it is spatially and temporally variable, and the variability is not solely related to interactions with the solar environment. Sources for the known exospheric species include impact vaporization, ion sputtering, thermal and photon stimulated desorption, crustal outgassing, and neutralization of solar wind ions. The relative importance of these production mechanisms has been debated, but they

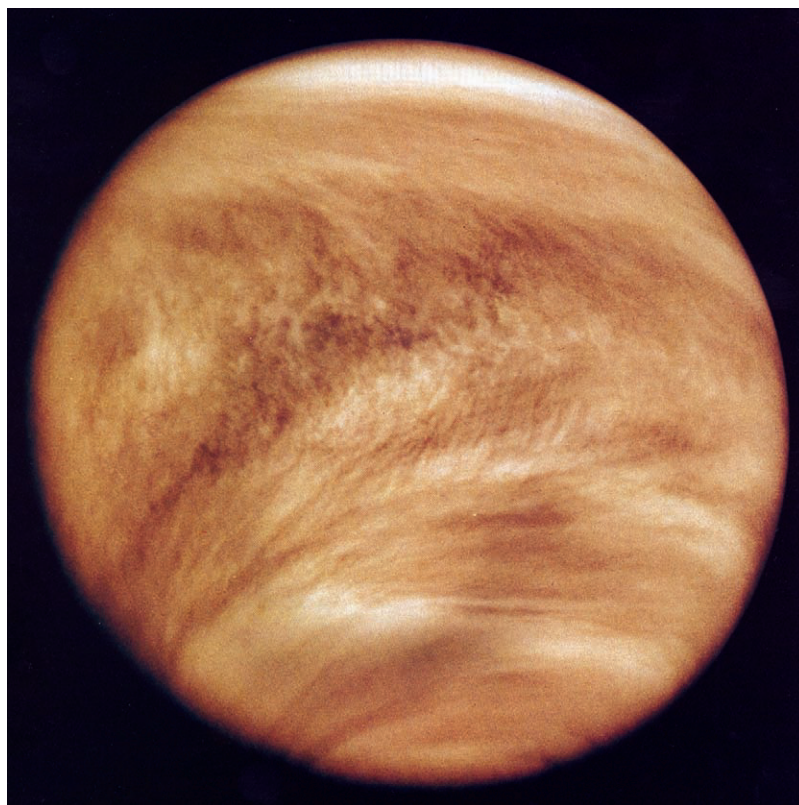
predict the existence of several species (such as Ar, Si, Al, Mg, Fe, S, and OH) that have yet to be detected. The *MESSENGER* spacecraft carries an ultraviolet spectrometer as part of the Mercury Atmosphere and Surface Composition Spectrometer (MASCS) instrument package. This spectrometer operates from 1150 to 6000 Å and its goal is to map the constituents of the atmosphere and provide information to relate them to specific source and production mechanisms.

The known lunar atmospheric species present in detectable abundances are Ar, He, Na, and K. So far, only upper limits on other species have been set using UV wavelengths. In the UV, the lunar atmosphere was initially studied at FUV wavelengths by the *Apollo 17* UVS, which provided upper limits on the number density of H, H<sub>2</sub>, O, C, N and CO. More recently, *HST* FOS NUV observations of the region away from the surface of the Moon resulted in upper limits on OH, Al, Si, and Mg abundances.

### 3.2 Venus

For more than half a century, the very dense Venus atmosphere has been known to be composed principally of carbon dioxide (CO<sub>2</sub>) based on the existence of strong spectral absorption features in the near-infrared spectrum. Several layers of clouds many kilometers thick composed of sulfuric acid completely cloak the surface. Although these clouds obscure the surface at visual and ultraviolet wavelengths, the *Magellan* spacecraft used radar to construct maps of the volcanic surface. Atmospheric measurements in the UV have been performed by sounding rockets and spacecraft, including *IUE*, the *Pioneer Venus* orbiter, and *HUT*. An image from *Pioneer Venus* is shown in Fig. 1. (This 3650 Å image is just outside the strict definition of the NUV range.) Ultraviolet images of Venus' atmosphere show distinctive cloud patterns; in particular, a horizontal "Y"-shaped cloud feature (discovered by *Mariner 10* Venus scientists in 1974) is visible near the equator. This feature may suggest atmospheric waves, analogous to high and low pressure cells on Earth. Bright clouds toward Venus' poles appear to follow latitude lines. The polar regions are bright, possibly showing a haze of small particles overlying the main clouds. The dark regions show the location of enhanced sulfur dioxide near the cloud tops.

Within a few years of launch, *IUE* identified several important trace constituents, including nitric oxide (NO), and confirmed the presence of several others, such as sulfur dioxide (SO<sub>2</sub>). The *Vega 1* and 2 probes measured local ultraviolet absorptions due primarily to SO<sub>2</sub> and aerosols. Ultraviolet reflectance spectra obtained during two sounding rocket observations in 1988 and 1991 found that SO<sub>2</sub> is the primary spectral absorber between 1900 and 2300 Å and that sulfur monoxide (SO) is also present in Venus' atmosphere. The EUV instrument aboard the *Galileo* spacecraft



**FIGURE 1** *Pioneer Venus* Orbiter Cloud Photopolarimeter (OCP) image of Venus at 3650 Å.

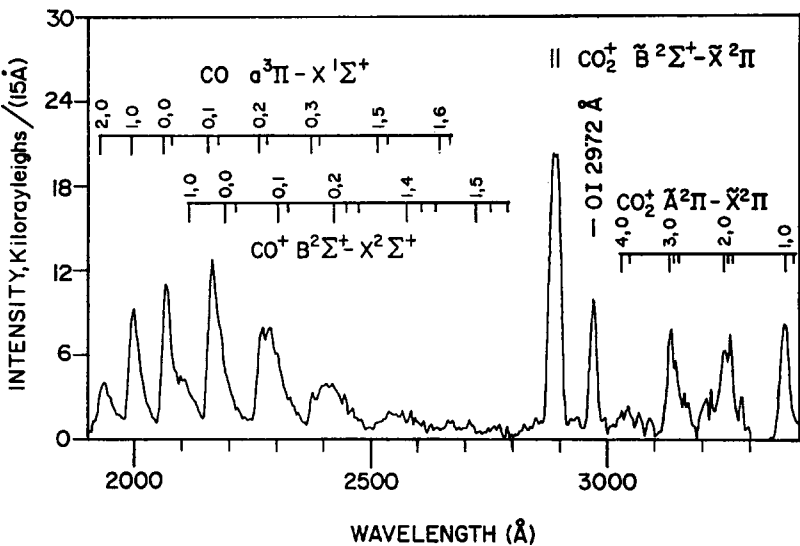
observed Venus in the extreme ultraviolet wavelength range (550–1250 Å) during its flyby. It detected emissions due to helium, ionized oxygen, atomic hydrogen, and an atomic hydrogen–atomic oxygen blend. In 1994, an extreme ultraviolet spectrograph (EUVS) was launched aboard a sounding rocket to observe the Venusian atmosphere from 825 to 110 Å. The EUVS identified several species, including N I, N II, N<sub>2</sub>, H I, O I, and O II. The results of the EUVS measurements are consistent with earlier observations by *IUE*, *Pioneer Venus*, *Venera 11* and *12*, and the *Galileo* EUV spectrometer. The *EUV* provided the first full EUV (70–760 Å) spectrum of Venus in 1998 and made brightness measurements on the He I (584 Å) and O II (539 Å) lines. The FUV spectrum of Venus is dominated by the CO Fourth-positive band system, as well as by neutral oxygen and carbon features, and has been measured by *HUT* (820–1840 Å) in 1995 and by *Cassini* UVIS.

*IUE* spectra of the Venus day side and night side obtained while Venus was near elongation displayed SO<sub>2</sub> absorptions at 2080–2180 Å, which when combined with the **column densities** reported by the *Pioneer Venus* orbiter and with ground-based observations, are a measure of the SO<sub>2</sub> **mixing ratio** with altitude and its variation at the top of the cloud deck. This provides information on its variation in spatial distribution and permits models to be constructed of the planet's atmospheric dynamics. Observations

of the Venus night side with *Pioneer Venus* orbiter and *IUE* detected the Venus nightglow, which is caused by the emission bands of nitric oxide (NO). Because of the short lifetime of NO on the night side, this finding implies the rapid day side–night side transport of material in the Venus atmosphere. Observations of the Venus day side have led to the discovery that the dayglow emission is carbon monoxide fluorescence, probably due to fluorescent scattering of solar Lyman-alpha radiation.

### 3.3 Mars

The atmosphere of Mars, like that of Venus, is dominated by carbon dioxide, and also consists of small amounts of N<sub>2</sub>, H<sub>2</sub>O, and their photochemical products. Mars' atmosphere is much less dense than Venus's atmosphere and is relatively transparent at most wavelengths. Therefore, ultraviolet to infrared observations of Mars reveal information about both its atmosphere and its surface. The observations of Mars by the UVS instruments on *Mariner 6* and *7* were the first to reveal the ultraviolet dayglow of that planet; later observations by *Mariner 9* confirmed and extended these results. NUV spectra (Fig. 2a) revealed the presence of CO (a-X) Cameron bands, CO<sup>+</sup> (B-X), CO<sub>2</sub><sup>+</sup> (A-X), and CO<sub>2</sub><sup>+</sup> (B-X) features. FUV spectra displayed oxygen features at 1304 and 1356 Å, neutral carbon emission features at 1561 and 1657



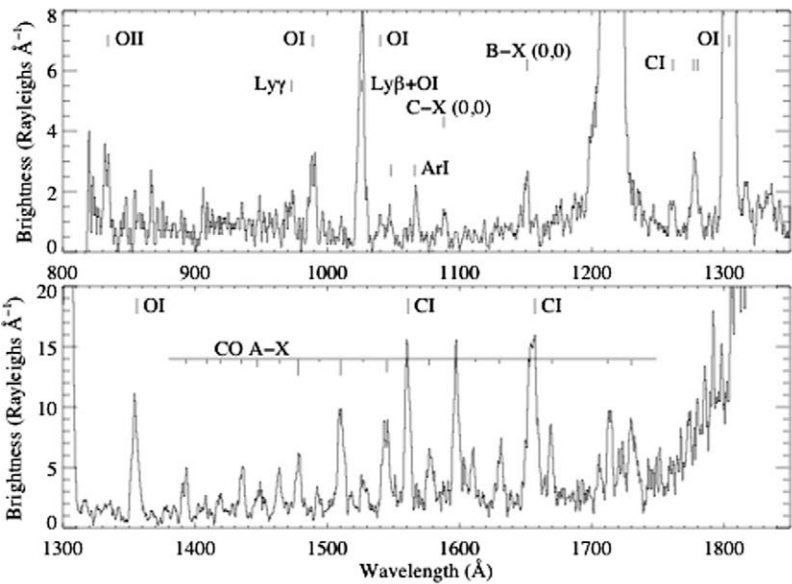
**FIGURE 2a** Mars dayglow as measured by the *Mariner 9* UVS. (Figure reproduced with permission from Elsevier.)

Å, and the CO (A-X) at fourth-positive bands. All these UV features of the martian airglow are products of processes involving Mars' CO<sub>2</sub> atmosphere. The CO<sub>2</sub><sup>+</sup> band systems are the result of a combination of photoionization excitation of CO<sub>2</sub> and fluorescent scattering of CO<sub>2</sub><sup>+</sup>, and the CO (a-X) and (A-X) bands are due to photon or electron dissociative excitation of CO<sub>2</sub>. The presence of escaping hydrogen (Ly-α at 1216 Å), suggested atomic hydrogen within the atmosphere, and also suggested the accumulation of oxygen and loss of water.

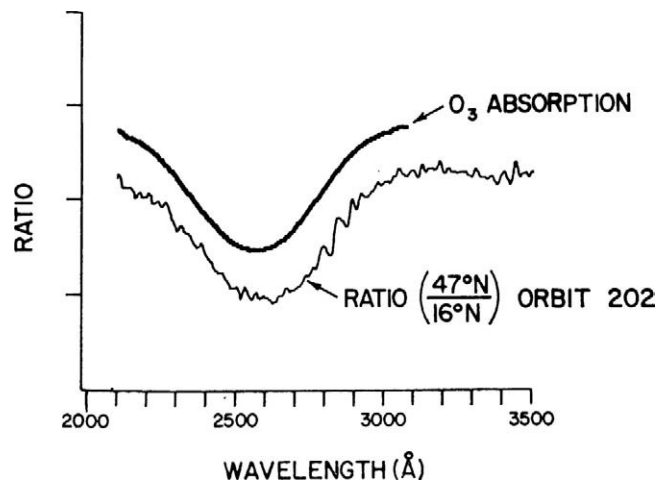
More recent (1995) *HUT* observations (820–1840 Å) confirmed these early *Mariner* results (Fig. 2b), and *FUSE* measurements detected molecular hydrogen (H<sub>2</sub>) emission features at 1070 and 1167 Å for the first time. *EUVE*

provided the first measurements of helium (584 Å) within the martian atmosphere. These helium observations have been used to set constraints on outgassing processes. Recent UV observations from the *Mars Express* spacecraft have made the first detection of the martian nightglow, revealing nitric oxide emission features similar to those seen on Venus, auroral emissions associated with crustal magnetic field features, and high altitude CO<sub>2</sub> ice clouds.

*Mariner 6* and *7* UVS observations revealed the presence of the Hartley band of ozone (O<sub>3</sub>), centered near 2550 Å. The feature was detected at the south polar cap, through ratios of south polar spectra to low latitude spectra (Fig. 2c). Further observations with the UVS on the *Mariner 9* orbiter revealed that the north and south polar ozone amount



**FIGURE 2b** Mars spectrum at EUV-FUV wavelengths as measured by *HUT*. (Figure reproduced with permission from AAS Publications.)



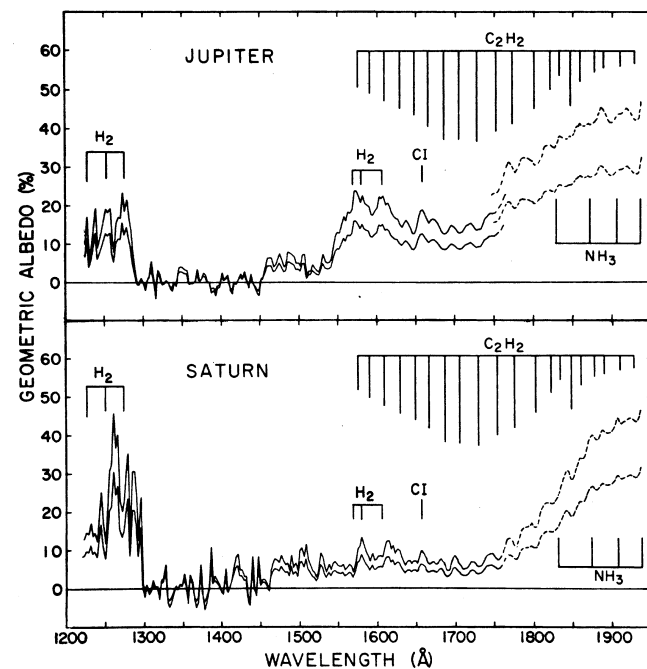
**FIGURE 2c** Ozone at high latitudes on Mars as measured by *Mariner 9* UVS. A high-latitude spectrum is shown ratioed to a low-latitude spectrum, compared with a laboratory spectrum of the Hartley band of  $O_3$ . (Figure reproduced with permission from Elsevier.)

varied with season. Ozone densities were highest in winter and lowest in summer, anticorrelated with atmospheric water vapor content. The correlation between higher amounts of ozone with a cold, clean, dry atmosphere led to the conclusion that ozone is formed through the combination of atomic and molecular oxygen, both of which are more readily present when less water is available. Subsequent *HST* observations have studied the seasonal variation of atmospheric ozone at low latitudes and have linked low latitude ozone abundance variations across the martian perihelion–aphelion cycle with the large annual water vapor variation due to the eccentricity of Mars’ orbit. *HST* observations have also been used to study atmospheric aerosol (dust and cloud) opacities. These UV observations demonstrate the critical function that is performed by the small amounts of  $H_2O$  in the martian atmosphere, which control the buildup of  $CO$  and  $O_2$ , and sustain the stability of  $CO_2$ .

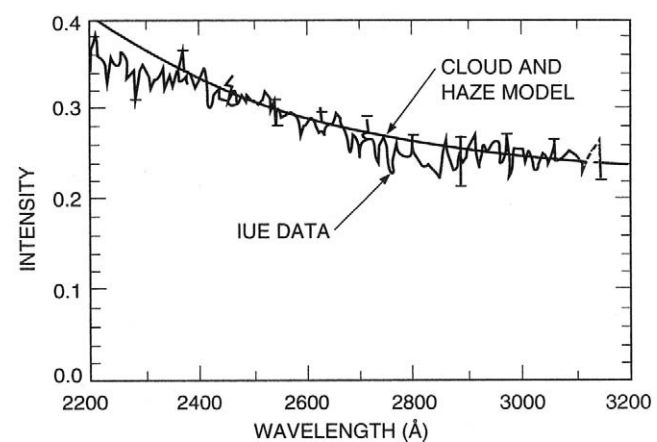
### 3.4 Jupiter

Jupiter, the target of numerous Earth-based observations as well as spacecraft flybys, is composed of 90% hydrogen and 10% helium, with small amounts of ammonia and methane. The uppermost layers of the atmosphere are observable in the ultraviolet wavelength range and display products of photochemical processes.

The first FUV spectra of Jupiter were measured in sounding rocket experiments in the late 1960s and early 1970s. These early measurements displayed  $H_2$  Lyman and Werner band emissions and hinted at the presence of absorption features due to  $C_2H_2$ ,  $C_2H_4$  and  $NH_3$ . Stellar occultation observations by the *Voyager* UVS were significant in providing measurements of upper atmospheric



**FIGURE 3a** FUV geometric albedos of Jupiter and Saturn as measured by *IUE*. The albedos are derived from composite spectra of the planets between 1978 and 1980. The upper and lower curves for each planet correspond to assumptions of uniform and cosine-limb-darkened disks, respectively. The dashed lines represent data that are uncertain in magnitude owing to the subtraction of scattered light and are regarded as upper limits. (Figure reproduced with permission from AAS Publications.)



**FIGURE 3b** The spectral geometric albedo of Jupiter as measured by *IUE*. The smooth solid line is the best fit from a model that assumes a layer of haze particles with single-scattering albedo of 0.42 that overlie a cloud deck with geometric albedo of 0.25. (Figure reproduced with permission from Elsevier.)



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temperatures. Early *IUE* observations confirmed that  $C_2H_2$  absorption bands are the dominant features in the 1650–1850 Å wavelength region. Figure 3a shows the geometric albedo of Jupiter in the FUV wavelength range, displaying  $C_2H_2$  and  $NH_3$  features, derived using a composite spectrum of Jupiter from the 1978–1980 time frame.

The spectral geometric albedo of Jupiter as measured by *IUE* at NUV wavelengths is shown in Fig. 3b. Most of this spectral behavior is attributable to hazes that are high above the cloud deck. The best-fit model (solid line in the figure) to the data occurs for a jovian cloud deck with a geometric albedo of 0.25 and for a haze composed of particles with a single scattering albedo of 0.42. Though such a result may not be able to provide an unambiguous identification of the materials that compose the haze, it can constrain the eligible candidate materials that are suggested by other observations.

*IUE* observations also permitted an ammonia–hydrogen mixing ratio to be calculated, and it was found to be  $5 \times 10^{-7}$ . The fact that *IUE* was able to observe the absorption features of these species indicates that they are above the jovian tropopause, where the clouds create an opaque barrier to light emitted from the material underneath and hence make spectral identification of the underlying material impossible. In July 1994, the comet Shoemaker–Levy 9 (SL-9) collided with Jupiter. It was not until the impact of the fragmented comet that studies of this underlying material became possible. The *EUVE* satellite observed Jupiter before, during, and after this event. *EUVE* found that 2 to 4 hours after the impact of several of the larger fragments, the amount of neutral helium temporarily increased by a factor of  $\sim 10$ . This transient increase is attributed to the interaction of sunlight with the widespread high-altitude remnants of the plumes from the larger impacts. *HST* also observed this event with the GHRS and the Faint-Object Camera (FOC). The ultraviolet spectra obtained by *HST* of Jupiter after the collision of SL-9 identified approximately 10 species of molecules and atoms in the perturbed atmosphere, many of which had never been detected before in Jupiter's atmosphere. Among these were  $S_2$ ,  $CS_2$ ,  $CS$ ,  $H_2S$ , and  $S^+$ , which are believed to be derived from a sulfur-bearing parent molecule native to Jupiter. The observations also detected stratospheric ammonia ( $NH_3$ ). Neutral and ionized metals, including Mg II, Mg I, Si I, Fe I, and Fe II, were also observed in emission and are believed to be from the SL-9 comet fragments. The surprising observation was the absence of absorptions due to oxygen-containing molecules.

A major focus of study at UV wavelengths is the polar regions of Jupiter and their impressive exhibit of auroral activity. Jupiter's auroral displays are the most energetic in the solar system. FUV measurements were first made using the *Voyager* UVS, and subsequent observations were performed by *IUE*. The far-ultraviolet emissions are dominated by the hydrogen Lyman-alpha and the  $H_2$  Lyman

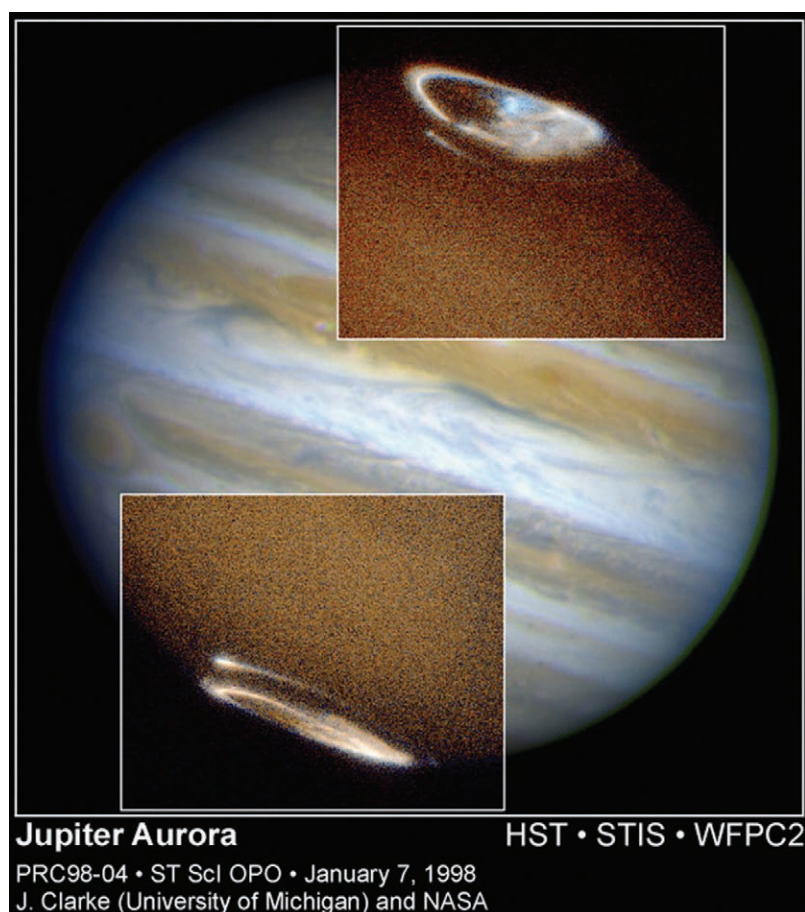
and Werner system bands. Synoptic observations of these ultraviolet emissions using *IUE* have shown that they vary with Jupiter's magnetic (not planetary) longitude, and hence these emissions are magnetospheric phenomena. *IUE* observations have been used to construct a spatial map of the Lyman-alpha emission and the data indicate that the emitting material is upwelling at about 50 m/s relative to the surrounding material. More intensive ultraviolet observations with *FUSE* and *HST* instruments GHRS, FOC, STIS, and ACS have measured the temporal variability within the aurora and temperature variations within the auroral ovals seen at both poles. These variations are reflections of possible distortions in the magnetic field of Jupiter. *HST* measured the first detection of reversed Lyman-alpha emissions, which are linked to variable atomic hydrogen. Estimates of vertical column densities ( $1\text{--}5 \times 10^{16} \text{ cm}^{-2}$ ) of atomic hydrogen above the auroral source have been made. *HST* has also detected ultraviolet emission from a superthermal hydrogen population. The *Galileo* spacecraft EUV and UVS spectrometers also observed Jupiter's aurora. These observations have placed constraints on the vertical distribution of methane ( $CH_4$ ) in Jupiter's atmosphere. Slant methane column abundances are estimated to be  $2 \times 10^{16} \text{ cm}^{-2}$  in the north and  $5 \times 10^{16} \text{ cm}^{-2}$  in the south based on the *Galileo* observations. *Cassini* UVIS measurements showed that Jupiter's aurora responded strongly to the compression events produced when large solar coronal mass ejections reached Jupiter's magnetosphere. Figure 4 displays Jupiter's UV aurorae as imaged by *HST* in 1998. Evident in Fig. 4 is the auroral “footprint” of Io, where the field line intersecting Io connects to the planet, revealing the magnetospheric relationship between the planet and the moon. Magnetic footprints of the other moons also exist but are less obvious in this image.

Bright H Ly- $\alpha$  emissions have been observed from Jupiter's equatorial region, and the source of this “equatorial bulge” has been debated. The source is likely a combination of charged particle excitation and solar resonance scattering and fluorescence. The emission has been shown to be consistent with resonant scattering of solar Ly- $\alpha$  with a large planetary line width, requiring a fractional ( $\sim 1\%$ ) suprathermal population of fast H atoms in the uppermost atmosphere. The fast atoms are likely due to dissociative excitation of molecular hydrogen ( $H_2$ ).

### 3.5 Saturn

Like Jupiter, Saturn's atmosphere is dominated by hydrogen and helium, with traces of water, ammonia, and methane. The far-ultraviolet spectrum of Saturn was first measured in sounding rocket experiments in 1978. Absorption features in the ultraviolet spectrum of Saturn that have been associated with acetylene ( $C_2H_2$ ) in the upper atmosphere were discovered using early *IUE* measurements. Figure 3a displays the FUV geometric albedo of Saturn derived using





**FIGURE 4** *HST* image of Jupiter aurora. The magnetic “footprint” of Io, marking the location where magnetic field lines joining the moon and Jupiter connect with the planet, is also seen as a bright spot with a tail outside the main auroral oval. (Image credit: J. Clarke, NASA.)

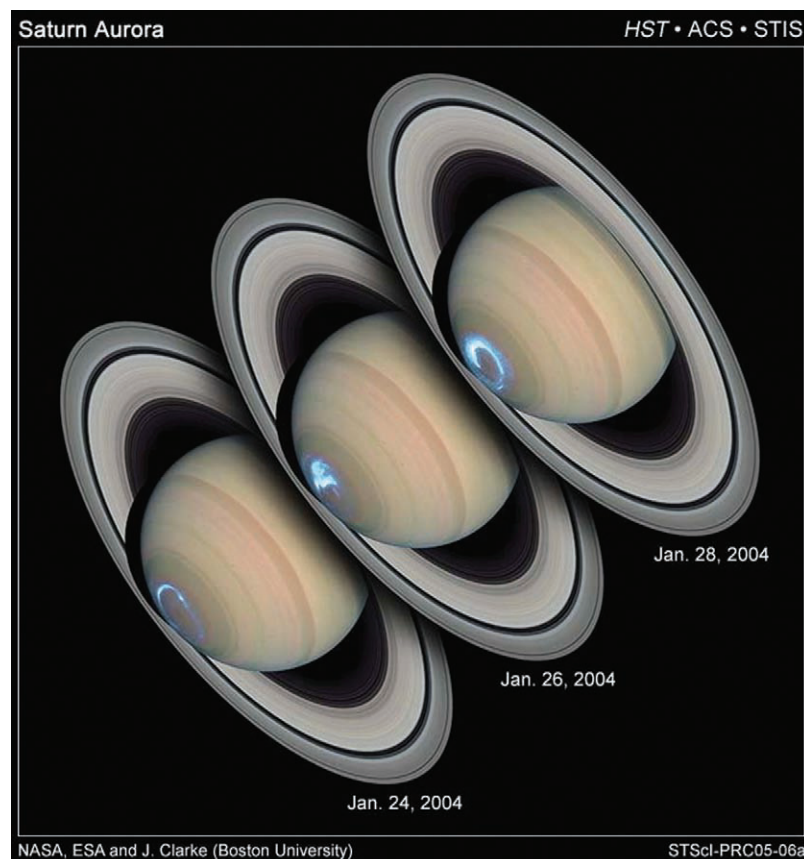
a composite of *IUE* Saturn spectra from the 1978–1980 time period. The mixing ratio of the acetylene is about  $1 \times 10^{-7}$ . Although acetylene is a well-known strong absorber of ultraviolet radiation, it alone cannot explain the low UV spectral geometric albedo of Saturn that has been reported by *IUE*. Other ultraviolet-absorbing materials must be present. Comparisons of laboratory spectra of  $C_2H_2$ ,  $PH_3$ ,  $AsH_3$ , and  $GeH_4$  with the *IUE* observations show that the best-fit model for Saturn’s atmospheric ultraviolet spectrum includes absorptions by  $C_2H_2$ ,  $H_2O$ ,  $CH_4$ ,  $C_2H_6$ ,  $PH_3$ , and  $GeH_4$ . The distribution of  $PH_3$  and  $GeH_4$  decreases with increasing altitude in these models, suggesting that ultraviolet photolysis is an important process occurring at higher altitudes.

Pole-to-pole mapping studies of the hydrogen Lyman-alpha emission across Saturn’s disk led to the discovery of pronounced spatial asymmetries in the emission. Other observations of hydrogen do not find a variation in intensity with rotational period as with Jupiter. There is no rotational bulge in the Lyman-alpha emission as seen on Jupiter. This is probably due to the fact that Saturn’s magnetic pole is coincident with the rotational pole, whereas in Jupiter’s case the poles are offset.

Like Jupiter, Saturn displays auroral activity. On both planets this auroral activity also creates aerosols that are detectable in the ultraviolet as dark-absorbing regions. *HST* FOC ultraviolet observations discovered a dark oval encircling the north magnetic pole that is spatially coincident with the aurora detected by the *Voyager* UVS. *Voyager 2* ultraviolet PPS measurements also demonstrate a geographical correlation between the auroral zones of Jupiter and Saturn with UV-dark polar regions. Additional ultraviolet observations with the *HST* FOC of Saturn’s northern ultraviolet aurora and polar haze support the hypothesis that the polar haze particles are composed of hydrocarbon aerosols produced during  $H_2^+$  auroral activity. More recent *HST* UV imaging of Saturn’s aurorae (Fig. 5) shows that they behave differently from Jupiter’s aurorae, varying in brightness and shifting in latitude.

### 3.6 Uranus

Uranus presents a unique observational circumstance to the inner solar system observer because of the fact that its pole is inclined  $89^\circ$  to the ecliptic and that at the present position in its 84 year orbit about the Sun it presents its pole



**FIGURE 5** *HST* images of Saturn's varying aurorae. (Image credit: J. Clarke, NASA.)

to Earth. This unusual inclination, combined with its great distance from Earth, makes it impossible to use an Earth-based instrument to undertake pole-to-pole comparisons as was done with Jupiter and Saturn. Uranus has a geometric albedo at NUV wavelengths of about 0.5, more than twice that of Jupiter and Saturn. This suggests that additional absorbers are present in the jovian and Saturnian atmospheres that are not present in the atmosphere of Uranus. Both Uranus and Neptune possess hot thermospheres and stratospheres that are substantially clear of hydrocarbons and other heavy constituents, making the UV albedos higher than for Jupiter and Saturn. A sharp increase in measured reflectance intensity at wavelengths longward of 1500 Å is indicative of acetylene ( $C_2H_2$ ) present in the atmosphere of Uranus.

*Voyager 2* spacecraft observations of Uranus found a very small internal heat source compared to the large internal heat sources found at Jupiter and Saturn. This suggests that there is very little atmospheric mixing driven by heating and buoyancy in the Uranian atmosphere. Thus, ultraviolet observations are able to sense a deeper region of the atmosphere.

The ultraviolet emissions from Uranus' atmosphere have been measured by *IUE* and the *Voyager UVS*. To increase the signal-to-noise ratio, *IUE* observers used principally

low-resolution observations and binned broad-wavelength regions together to search for broadband absorbers at ultraviolet wavelengths. Analysis of the *IUE* observations detected acetylene absorptions, which were also detected on Jupiter and Saturn. Based on these observations, the mixing ratio of the acetylene is estimated to be  $3 \times 10^{-8}$ . Analysis of the *Voyager UVS* observations of  $H_2$  band ultraviolet air-glow emissions shows aurora at both magnetic poles, which are offset from the rotational poles by  $\sim 60^\circ$ . The auroral emissions on Uranus are very localized in magnetic longitude and do not form complete auroral ovals as are seen on Jupiter and Saturn.

### 3.7 Neptune

Neptune is so distant that only broadband ultraviolet measurements are possible from Earth orbit. The geometric albedo of Neptune measured by *IUE* is 0.5, which, like that of Uranus, is twice that of Jupiter and Saturn. Below 1500 Å, Neptune's albedo is reduced by the higher hydrocarbon abundance carried into its stratosphere by its more vigorous vertical transport. Most of the important data for Neptune at ultraviolet wavelengths have come from the UVS onboard the *Voyager 2* spacecraft and from *HST*.  $CH_4$  and  $C_2H_6$  abundances inferred from the *Voyager UVS* solar

occultation experiment are between 0.0006 and 0.005 mole fraction for  $\text{CH}_4$  in the lower stratosphere (with a mixing ratio of  $5\text{--}100 \times 10^{-5}$ ) and a density of  $\text{C}_2\text{H}_6$  estimated to reach  $3 \times 10^9 \text{ cm}^{-3}$ .

In 1994, *HST* imaged Neptune in six broadband filters, one of which was in the ultraviolet. The goal of these observations was to study the cloud structure on Neptune and compare the measurements with the observations made by *Voyager 2*. The *HST* images showed that the Great Dark Spot seen by *Voyager* no longer existed, but a new large dark feature of comparable size had appeared in the northern latitudes. *HST* measurements also detected weak carbon monoxide lines at 1992 and 2063 Å, suggesting a mixing ratio of  $\sim 3 \pm 2 \times 10^{-6}$  in the upper troposphere.

*Voyager 2* UVS measurements tentatively identified weak auroral emissions at Neptune's South Pole, interpreted as  $\text{H}_2$  emissions. The North Pole was not observed by *Voyager*, so it is unknown whether that hemisphere displays aurora.

### 3.8 Pluto

Pluto and its large satellite Charon are at a great distance from Earth and are quite small compared to the four gas-giant planets that populate the outer solar system. UV spectroscopy is a potentially rich source of information about these icy bodies due to the atmospheric chemistry that is likely occurring. The presence of methane in Pluto's atmosphere suggests that photochemical products should include hydrocarbons and nitriles, detectable at UV wavelengths. *IUE* obtained a few spectra of these objects and observed that the ultraviolet albedos vary with rotation. The amplitude of the rotational variation as measured at ultraviolet wavelengths by *IUE* is greater than the rotational variation measured at longer wavelengths by Earth-based observers, consistent with the presence of an absorbing material that is spectrally active in the 3200–4800 Å wavelength range; the geometric albedo of Pluto in the NUV is spectrally flat. The composition of the absorbing material is unknown. [See PLUTO.]

Observations made with the FOS on *HST* were used to determine upper limits on Pluto's predicted atmospheric species  $\text{C}_4\text{H}_2$ ,  $\text{C}_6\text{H}_2$ ,  $\text{HC}_3\text{N}$ , and  $\text{C}_4\text{N}_2$  of  $1.6 \times 10^{16}$ ,  $1.8 \times 10^{16}$ ,  $2.7 \times 10^{16}$ , and  $4 \times 10^{16} \text{ cm}^{-2}$ , respectively. The ultraviolet spectrum of Pluto's satellite, Charon, was also measured by the *HST* FOS and was found to have a spectrally flat geometric albedo in the NUV; Charon's spectrum does not exhibit any absorption or emission features that provide compositional clues.

The Pluto–Charon system is the target of the upcoming *New Horizons* flyby mission, which will include an FUV imaging spectrograph (which operates from 520 to 1870 Å) to probe the atmospheres and surfaces of these distant worlds.

### 3.9 Galilean Satellites

Jupiter's moon Io has one of the most unique atmospheres in the solar system: The primary sources of the atmosphere are volcanic emissions and sublimation of  $\text{SO}_2$  frost on the surface. The result is a tenuous, patchy atmosphere made up of  $\text{SO}_2$ ,  $\text{SO}$ ,  $\text{S}_2$ ,  $\text{S}$ , and  $\text{O}$ ; trace species include  $\text{Na}$ ,  $\text{K}$ ,  $\text{Cl}$ ,  $\text{NaCl}$ , and  $\text{H}$ . Gaseous  $\text{SO}_2$  can be deposited onto the surface at night or during eclipse by Jupiter. Material is also lost to the torus as the ionized material sweeps by Io. Gaseous  $\text{SO}_2$  was discovered at Io by the IRIS instrument on *Voyager*; since that discovery, much study of Io's atmospheric processes has been made at UV wavelengths. The *IUE*, *HST*, and the *Galileo* UVS have made measurements of Io at near-UV wavelengths (2000–3500 Å). Far-ultraviolet observations from *HST* and *Cassini* UVIS identified emissions from neutral oxygen and sulfur and have been used in mapping the distribution of the  $\text{SO}_2$  atmosphere.

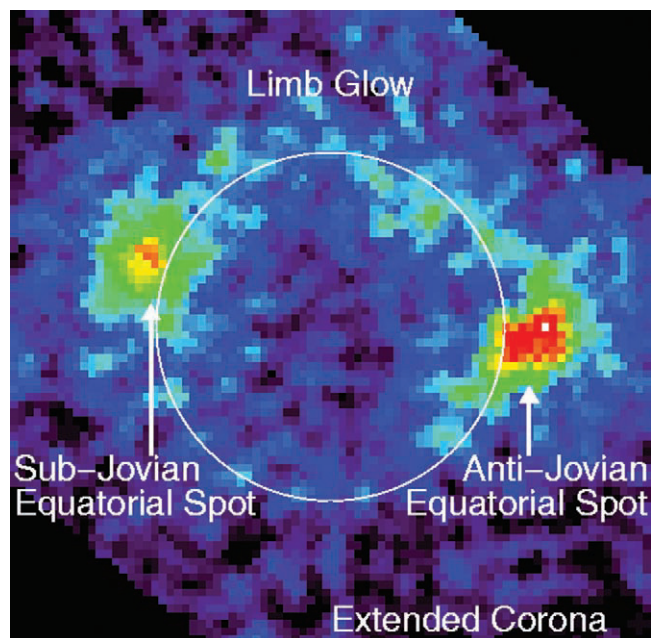
Associated with Io is a plasma torus, or a donut-shaped ion cloud centered at Io's orbital radius. This torus has been studied by *Pioneer*, *Voyager*, *IUE*, *HST*, *EUVE*, *HUT*, *FUSE*, *Galileo*, and *Cassini*. Oxygen, sulfur, and sodium ions are the major constituents of the torus, and protons are present at  $\sim 10\%$  abundance; chlorine ions have also been detected. The torus is not uniform, and the density of ions shows various asymmetries dependent on Io's position and dawn–dusk timings, in addition to temporal variations.

Intriguing auroral features are a consequence of Io's  $\text{SO}_2$  atmosphere, resulting from electron impact excitation of atomic oxygen and sulfur, and electron dissociation and excitation of  $\text{SO}_2$ , and have been observed at visible and FUV wavelengths. The Io flux tubes (IFT) and the Io plasma torus are the two primary sources of electrons in the Io environment. Due to the  $10^\circ$  tilt of Jupiter's magnetic field, Io is alternately above and below the magnetic equator (depending upon Io's System III jovian magnetic longitude,  $\lambda_{\text{III}}$ ), the primary region of the torus electrons. Furthermore, the tangent points between field-aligned electrons and Io's atmosphere change as Jupiter rotates. The interaction of the torus electrons and the IFT electrons with Io's atmosphere has been detected at FUV wavelengths. Equatorial spots (Fig. 6) have been observed to wobble up and down, reflecting the changing location of the IFT tangent points in time. The equatorial spot on the antijovian hemisphere has been measured to be brighter than the spot on the subjovian hemisphere, likely due to the motion of electrons through Io's atmosphere by the Hall effect, with hotter electrons on the antijovian side. [See IO: THE VOLCANIC MOON.]

Observations with the *HST* GHRS have detected atomic oxygen emissions at 1304 and 1356 Å from Jupiter's satellite Europa, which have been interpreted as evidence for a tenuous  $\text{O}_2$  atmosphere about this satellite. The source of this oxygen atmosphere is likely sputtering of the icy surface by corotating magnetospheric particles. These emission

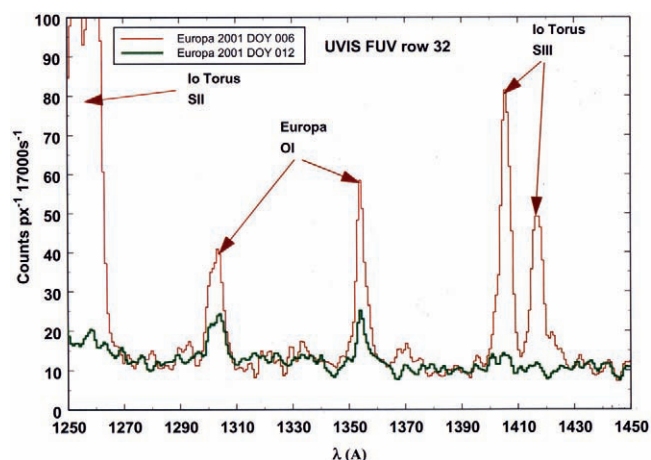


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**FIGURE 6** Io imaged at 1356 Å by *HST* STIS. Equatorial “spots” are the result of interaction between electrons flowing along the Io flux tubes with Io’s SO<sub>2</sub> atmosphere. (Figure courtesy of K. Retherford.)

features were also measured by the *Cassini* UVIS instrument during the Jupiter system flyby in 1999–2000 (Fig. 7). Similar oxygen emission features were detected by *HST* at Ganymede, though it was found that Ganymede’s emissions are restricted primarily to the polar regions. The emis-



**FIGURE 7** The FUV spectrum of Europa as measured by *Cassini* UVIS on January 6, 2001, and January 12, 2001. Neutral oxygen emission features appear at 1304 and 1356 Å in both observations. Io torus emission features from ionized sulfur also appear. The presence of the O I features is due to electron dissociation and excitation of a tenuous O<sub>2</sub> atmosphere at Europa. (Figure reproduced with permission from Elsevier.)

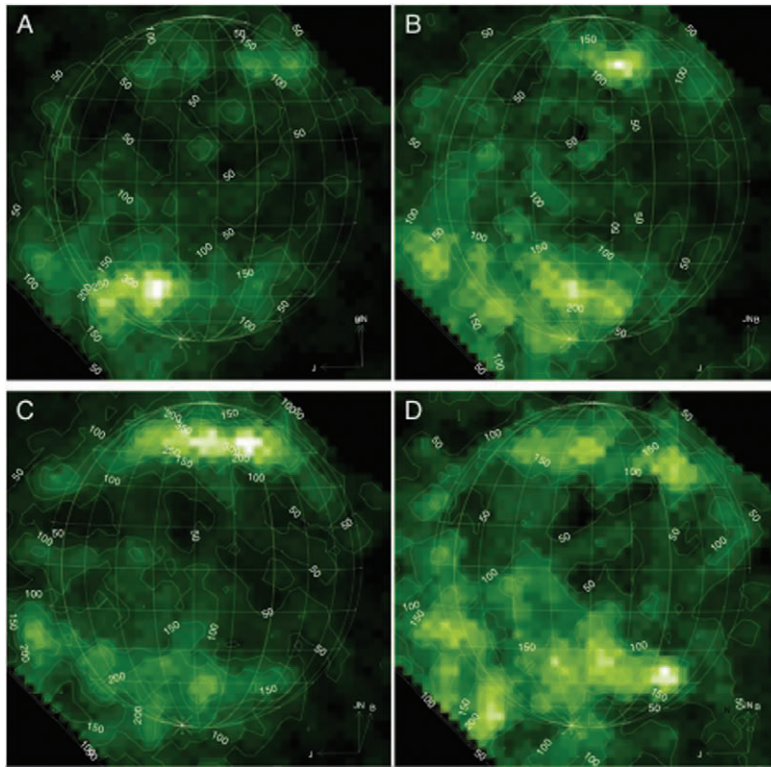
sions are auroral features produced by dissociative excitation of O<sub>2</sub> by electrons traveling along the field lines of Ganymede’s own magnetosphere. *HST* imaged Ganymede’s auroral emissions at 1356 Å and found them to be temporally and longitudinally variable (Fig. 8). *Galileo*’s UVS detected hydrogen escaping from Ganymede, possibly due to sputtering of Ganymede’s surface by charged particles. Callisto, in contrast to Ganymede and Europa, does not exhibit oxygen emission features. An analysis of *HST* measurements found that the oxygen and CO emission features, expected after the discovery of CO<sub>2</sub> gas, are so faint that Callisto’s interaction with the magnetosphere is like that of a unipolar inductor, and that another species such as O<sub>2</sub> is likely abundant, enhancing the ionosphere and its conductivity. Callisto’s ionosphere is apparently of sufficient conductivity to reduce the flow of plasma into its atmosphere and inhibits oxygen emission features in contrast to Ganymede and Europa.

### 3.10 Titan and Triton

Saturn’s satellite Titan and Neptune’s satellite Triton are among the largest satellites in the solar system. In addition, they are far from the Sun; therefore, the reduced solar energy allows the atmospheric gases to remain cold enough that they cannot easily escape by thermal processes. [See TITAN; TRITON.]

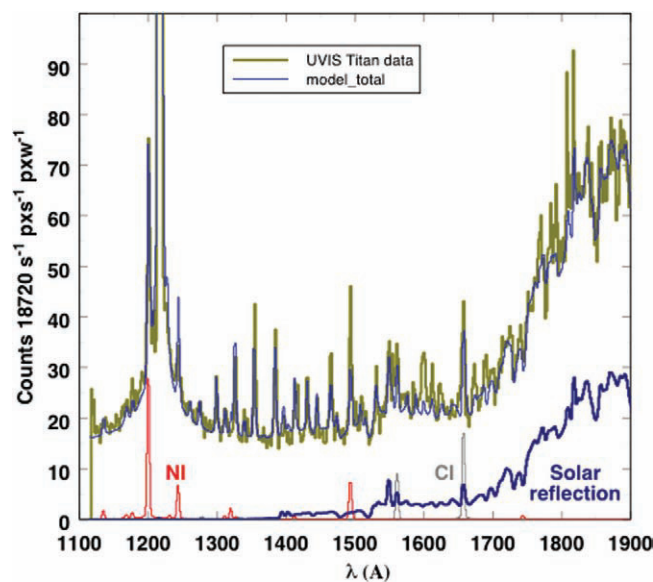
Titan is a solar system curiosity due to its very thick (~1.5 bar) nitrogen atmosphere, which prevents UV observations of the surface. Ground-based and *Voyager* spacecraft observations have identified methane (CH<sub>4</sub>) as a significant constituent of Titan’s atmosphere. Analyses of *IUE* and *HST* observations of Titan at NUV wavelengths have placed constraints on the properties of Titan’s high-altitude haze and the abundances of simple organic compounds such as acetylene (C<sub>2</sub>H<sub>2</sub>). At FUV wavelengths, observations by *Cassini* UVIS demonstrate the presence of molecular and atomic nitrogen based on emission features due to electron dissociation and excitation (Fig. 9).

Triton’s surface contains N<sub>2</sub>, CO, and CH<sub>4</sub> frosts, which are highly volatile and in a continual state of exchange between the atmosphere and surface. *IUE* observations of distant Triton likely tested the limits of *IUE*’s sensitivity. The photopolarimeter on *Voyager 2* measured an albedo of 0.59 on all sides of Triton. *HST* FOS observations in 1993 detected broad apparent absorption features centered near 2750 Å and between 2000 and 2100 Å. The FOS analysis also led to mixing ratio upper limits for atmospheric constituents of OH, NO, and CO of  $3 \times 10^{-6}$ ,  $8 \times 10^{-5}$ , and  $1.5 \times 10^{-2}$ , respectively. *HST* STIS observations from August to September 1999 showed that Triton’s albedo in the 2500–3200 Å range was 15–30% brighter, and also spectrally redder, than measured by the *HST* FOS in 1993, suggesting that Triton’s NUV albedo undergoes changes on timescales shorter than the seasonal cycle. Such changes



**FIGURE 8** *HST* STIS images of Ganymede's UV aurora. The images represent neutral oxygen at 1356 Å and reveal brightening in the polar regions that is variable over the four *HST* orbits during which the imaging took place. (Figure reproduced with permission from AAS Publications.)

may be due to bright frost deposition or to the emplacement of a relatively dark UV material. At this point, the source of the dramatic, short-timescale changes in UV brightness on Triton remains unknown.



**FIGURE 9** Titan's FUV spectrum as measured by the *Cassini* UVIS. The spectrum includes an overall continuum due to reflected solar light, in addition to emission features due to electron dissociation and excitation of molecular and atomic nitrogen. (Figure courtesy of D. Shemansky.)

#### 4. Observations of Solid Surfaces

Many solid-state materials that make up the surfaces of solar system objects exhibit spectral absorption features, and thus it is possible to identify or constrain the abundance of solid components on the surfaces of these objects. This is accomplished by comparing the spectral geometric albedo of the object with the reflection spectrum of the solid-state materials as measured in the laboratory. The following discussion focuses on ultraviolet observations of solid surfaces throughout the solar system. Light is reflected from particulate surfaces by volume scattering and surface scattering. At longer visible and near-IR wavelengths, volume scattering dominates in most materials. At shorter NUV and FUV wavelengths, surface scattering dominates. Absorptions in the NUV are generally due to charge-transfer and result in rather broad absorption features, compared with absorption features in the near-IR, usually due to weaker electronic transitions. Many non-ice materials are absorbing in the NUV, including organics, sulfur compounds, and many refractory materials. There is usually an absorption edge between 3000 and 3700 Å. In contrast, ices such as H<sub>2</sub>O and CO<sub>2</sub> are not very absorbing at visible wavelengths and at NUV wavelengths. (Water and CO<sub>2</sub> ice exhibit strong absorption features near 1650 Å.) Atmosphereless bodies in the solar system experience weathering, whereby their surfaces are affected by bombardment of micrometeorites as well as charged particles, either from the solar wind in the case of the Moon and asteroids or from the magnetosphere

| TABLE 1    Ultraviolet Geometric Albedos of the Galilean Satellites |               |               |               |
|---------------------------------------------------------------------|---------------|---------------|---------------|
|                                                                     | 2600–2700 Å   | 2800–3000 Å   | 3000–3200 Å   |
| Io (leading)                                                        | 0.015 ± 0.001 | 0.017 ± 0.001 | 0.042 ± 0.001 |
| Io (trailing)                                                       | 0.028 ± 0.002 | 0.030 ± 0.005 | 0.038 ± 0.003 |
| Europa (leading)                                                    | 0.213 ± 0.004 | 0.347 ± 0.010 | 0.407 ± 0.020 |
| Europa (trailing)                                                   | 0.118 ± 0.002 | 0.164 ± 0.004 | 0.222 ± 0.006 |
| Ganymede (leading)                                                  | 0.15 ± 0.007  | 0.190 ± 0.009 | 0.200 ± 0.001 |
| Ganymede (trailing)                                                 | 0.050 ± 0.003 | 0.060 ± 0.004 | 0.080 ± 0.008 |
| Callisto (leading)                                                  | 0.040 ± 0.008 | 0.049 ± 0.001 | 0.066 ± 0.002 |
| Callisto (trailing)                                                 | 0.056 ± 0.002 | 0.064 ± 0.002 | 0.105 ± 0.008 |

of the parent planet in the case of the icy satellites of the outer solar system. Particularly in the case of icy surfaces, radiation-aged surfaces tend to be darker at UV wavelengths. Thus, UV observations of icy surfaces can be used to indicate ice “freshness,” or amount of contamination.

4.1 Galilean Satellites

The first in-depth ultraviolet studies of the Galilean satellites (Io, Europa, Ganymede, and Callisto) were accomplished with the use of the *IUE* satellite. Subsequent disk-integrated observations with *HST* supported the initial findings of *IUE*, in addition to adding to our knowledge of the composition of the surfaces of these satellites. Galileo studies contributed to disk-resolved studies of these bodies.

The very high spatial resolution provided by *Voyager* and *Galileo* visible images shows that the Galilean satellites, particularly Io, are variegated in color on continental scales. Compositional information may be derived from high spectral resolution studies from Earth or near Earth orbit because the satellite’s synchronous rotation permits any given full-disk observation of a satellite to be associated with a uniquely defined hemisphere of that particular object. The extension of the available spectral range to shorter wavelengths with ultraviolet telescopes enhances this data set by permitting the identification of more absorption features, thereby providing further constraints on the compositional models that have been developed. [See PLANETARY SATELLITES.]

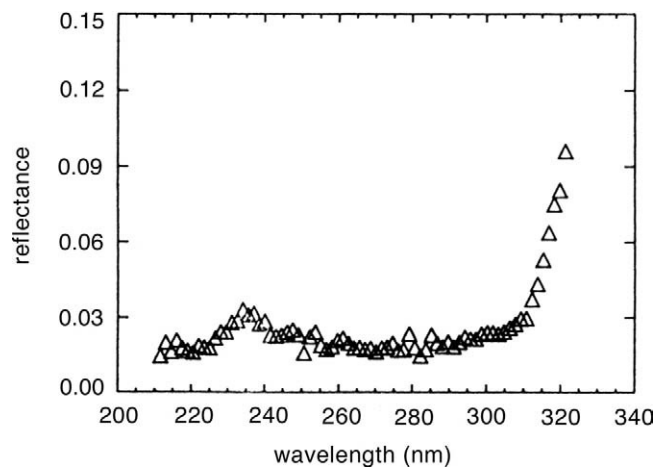
The Galilean satellites are phase-locked, so that one hemisphere (the subjovian, central longitude of 0° W) faces Jupiter at all times. The leading hemisphere is the side that faces the direction of motion of the satellite in its orbit and is centered on 90°W longitude, while the trailing hemisphere has a central longitude of 270°W. The corotating charged particles of Jupiter’s magnetosphere have orbital speeds greater than those of the moons, so that the plasma sweeps by the moons, impacting primarily the trailing sides. An in-depth study of several hundred *IUE* spectra of the Galilean satellites revealed significant hemispheric UV spectral asymmetries that are indicative of composi-

tional variations. Ratios of spectra from the leading and trailing hemispheres are a useful tool for studying hemispheric compositional variations.

Table 1 lists the geometric albedos of the Galilean satellites in three NUV wavelength bands, for the leading and trailing hemispheres. This table displays the significant hemispheric differences in brightness exhibited by these moons. Io’s leading hemisphere is brighter than the trailing hemisphere only in the longest NUV wavelength band. Shortward of 3000 Å, Io’s trailing side has a higher albedo than its leading side, just the opposite of what is seen when Io is observed at visible wavelengths. (*IUE*’s precursor, *OA0-2*, measured Io’s albedo at 2590 Å to be just 3%, in marked contrast to its 70% albedo at visible wavelength. This result was so unusual that it remained in doubt until *IUE* confirmed it by measuring Io’s spectrum in this spectral range.) Io’s reversal in brightness associated with orbital phase is more pronounced than for any other object in the solar system, and proved to be important in efforts to determine the surface composition variation in longitude across Io’s surface. It can be directly inferred from the Io data that there is a longitudinally asymmetric distribution of a spectrally active surface component on Io’s surface. The material was determined to be sulfur dioxide (SO<sub>2</sub>) frost, which is strongly absorbing shortward of ~3200 Å and very reflective longward of that wavelength, as a result of *IUE* observations. Sulfur dioxide frost is in greatest abundance on the leading hemisphere of Io, and it is in least abundance on the trailing hemisphere. Figure 10 shows a Galileo UVS spectrum that displays Io’s dramatic increase in albedo between 2000 and 3300 Å.

Europa and Ganymede exhibit a variation in brightness at NUV wavelengths with orbital phase that is in the same sense as the variation reported at the visible wavelengths; at all NUV wavelengths, these objects are brighter on their leading sides than on their trailing sides. A gradual decrease in albedo toward shorter wavelengths occurs on both hemispheres of both objects. The ratio of *IUE* spectra of Europa’s trailing hemisphere to its leading hemisphere led to the discovery of an absorption feature present primarily on the trailing hemisphere centered near 2800 Å. The absorption

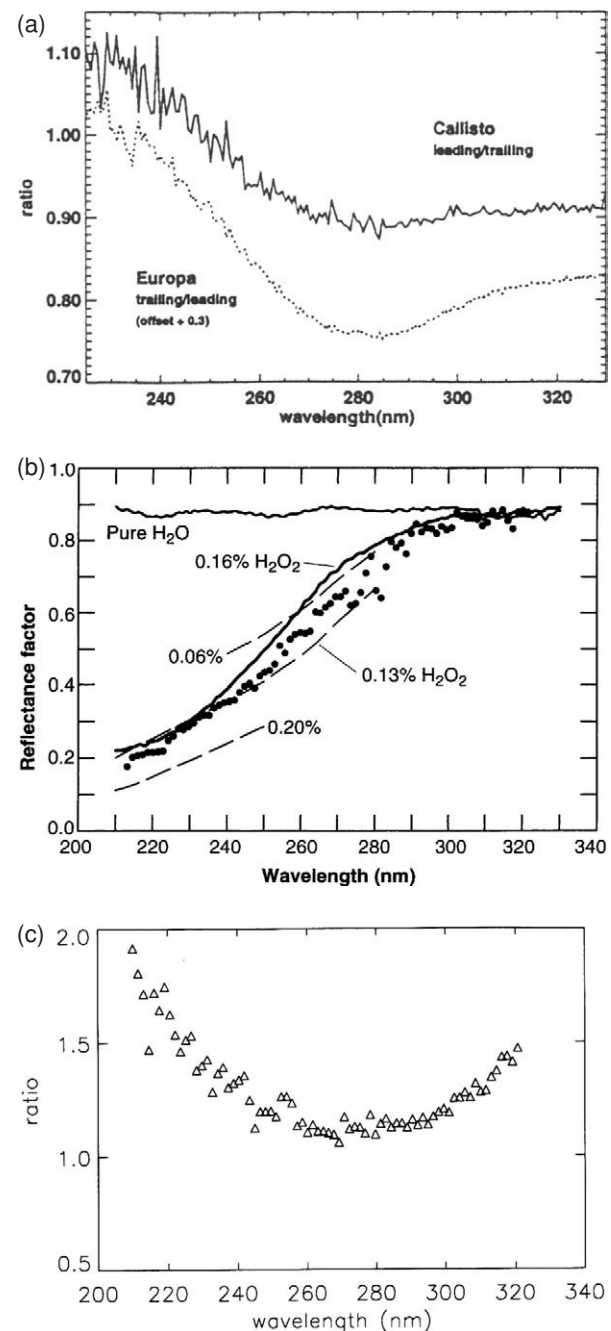




**FIGURE 10** The NUV reflectance spectrum of Io, displaying the dramatic increase in brightness with wavelength starting at  $\sim 320$  nm, due to the presence of  $\text{SO}_2$  frost on the surface. This spectrum was measured by the *Galileo* UVS. The original discovery of  $\text{SO}_2$  frost on the surface of Io in NUV spectra was made using *IUE* spectra. (Figure reproduced with permission from AAS Publications.)

feature, displayed in Fig. 11a, was attributed to an S—O bond and was suggested to be due to implantation of sulfur ions into the ice lattice on the trailing hemisphere. *HST* measurements confirmed the absorption feature, and it was suggested that the feature was similar to laboratory spectra of  $\text{SO}_2$  frost on water ice. Subsequent disk-resolved *Galileo* UVS measurements showed that the  $2800 \text{ \AA}$  absorption feature is strongest in regions associated with visibly dark terrain. These locations have also been found to have relatively high concentrations of non-ice material, interpreted to be hydrated sulfuric acid or hydrated salt minerals. An additional *Galileo* discovery was the presence of hydrogen peroxide ( $\text{H}_2\text{O}_2$ ) on Europa, primarily in regions of lower non-ice concentrations, such as on the leading hemisphere (Fig. 11b). A study of Europa spectra from the early *IUE* era (1978–1984) compared to the late *IUE* era (1995–1996) suggested a temporal variation in Europa's leading and antijovian hemisphere spectra that may be linked with variations in  $\text{H}_2\text{O}_2$  abundances as a result of temporal variability in the space environment.

*IUE* spectra of Ganymede's trailing hemisphere ratioed to the leading hemisphere revealed the presence of a possible absorption feature centered close to  $2600 \text{ \AA}$ , though the signal was approaching the *IUE* detection limits. It was suggested that ozone ( $\text{O}_3$ ) in the ice could explain the apparent absorption feature. Subsequent *HST* measurements confirmed the presence of the  $\text{O}_3$  absorption feature in the ice lattice on the trailing hemisphere. Disk-resolved observations of Ganymede from *Galileo* showed that the  $\text{O}_3$  feature was strongest in the polar regions, and at large solar zenith angles, suggesting a connection with the magnetic field lines, or with photolysis or ice temperatures. Figure 11c displays the  $\text{O}_3$  absorption feature as measured



**FIGURE 11** Spectra of significant absorption features on the icy Galilean satellites. (a)  $\text{SO}_2$  absorption features on Europa and Callisto obtained by ratioing the spectra of the trailing to leading hemisphere (Europa) and the spectra of the leading to trailing hemisphere (Callisto). (Figure reproduced with permission from AAS Publications.) (b) Hydrogen peroxide ( $\text{H}_2\text{O}_2$ ) absorption feature as measured on Europa by the *Galileo* UVS. Also shown are mixture models for varying amounts of  $\text{H}_2\text{O}_2$  in a water ice mixture, and the spectrum of pure  $\text{H}_2\text{O}$  ice. (Figure reproduced with permission from AAAS/Science.) (c) Ozone ( $\text{O}_3$ ) absorption feature as measured on Ganymede by *Galileo* UVS. Shown is the ratio of a spectrum from the north polar region to a region on the leading hemisphere. The broad absorption feature mimics  $\text{O}_3$  in water ice. (Figure reproduced with permission from AAS Publications.)

by the *Galileo* UVS. Galileo measurements also found that the UV absorption feature associated with H<sub>2</sub>O<sub>2</sub> is present also on Ganymede and is anticorrelated with the O<sub>3</sub> concentrations.

Ground-based observations of Callisto have found that its albedo varies with **orbital phase angle** in the opposite sense to that of Europa and Ganymede (i.e., its trailing side has a higher albedo than its leading side). This is also true at NUV wavelengths. The albedo of Callisto decreases shortward of 5500 Å and continues to decrease throughout the NUV. Its albedo at all wavelengths is lower than the albedo of Europa and Ganymede. Analysis of many of the *IUE* spectra, in addition to *HST* spectra, shows a broad, weak absorption at 2800 Å similar to that seen on Europa (Fig. 11a). These observations suggest the presence of SO<sub>2</sub> in a few leading hemisphere regions. The source of this SO<sub>2</sub> is not well understood and may be linked with implantation of neutral sulfur flowing outward from Io. The H<sub>2</sub>O<sub>2</sub> absorption feature seen by *Galileo* UVS at Europa and Ganymede is also seen at Callisto, though the absorption feature is weaker and its distribution is less obvious.

4.2 Saturnian Satellites

All of Saturn’s large and medium-sized satellites, like the Galilean satellites and Earth’s Moon, are in synchronous rotation (with the exception of Phoebe and Hyperion). At visual wavelengths, Tethys, Dione, and Rhea all have leading side albedos that are 10–20% higher than those of their trailing sides, which suggests that there are longitudinal differences in chemical/mineralogical abundance and/or composition in the optically active regoliths of these objects. The hemispheric albedo asymmetry of Iapetus at visual wavelengths is extremely large (the trailing side is brighter by a factor of 5). Infrared observations of the large satellites of Saturn have identified water ice as the principal absorbing species of the optically active surface of Mimas, Enceladus, Tethys, Dione, and Rhea and the trailing (bright) hemisphere of Iapetus. The leading (dark) hemisphere of Iapetus does not show spectral features consistent with water ice

and has an infrared spectrum that is nearly featureless. The albedo of water ice alone is too high for the surfaces of the satellites to be covered only by this material. Other materials must be present in varying amounts to explain the albedos of all the Saturnian satellites. In the case of the dark hemisphere of Iapetus, the darkening material is most probably the dominant specie on the surface. Observations at improved spectral resolution and extended spectral range are required to identify these absorbers on the surfaces of the Saturnian satellites. The suite of instruments on the *Cassini* spacecraft will likely make progress in determining the other species that characterize the surfaces of these moons.

A limited number of UV observations of the Saturnian satellites have been undertaken from Earth orbit. Mimas and Enceladus are too close to Saturn to obtain useful spectra from Earth: There is too much scattered light from Saturn. The next satellites, in order of distance from Saturn, are Tethys, Dione, Rhea, and Iapetus; these moons were first successfully measured at UV wavelengths by *IUE* and later observed by *HST*. The ultraviolet geometric albedos for the Saturnian satellites were calculated for the three UV wavelength bandpasses using a combination of *IUE* and *HST* data. These are shown in Table 2.

The UV albedo of Tethys (~60%) is the highest of the Saturnian satellites and is comparable to the high visual albedo reported by *Voyager* and ground-based visual observations. The leading side of Dione is ~30% brighter than its trailing side, similar to the brightness variation reported from ground-based visual wavelength observations. The leading side of Rhea is ~60% brighter than its trailing side at the NUV wavelengths. This is more than the ~20% observed from the ground at visual wavelengths and may signal the presence of a UV absorption feature focused on the trailing hemisphere.

The UV albedo of Iapetus is consistent with the albedos reported at longer wavelengths from ground-based observations and the *Voyager* spacecraft. In the UV, as in the visual, the leading side of Iapetus is extremely absorbing, and the trailing side is comparable to the trailing side albedos

TABLE 2    Ultraviolet Geometric Albedos of the Saturnian Satellites

|                    | 2400–2700 Å  | 2800–3000 Å  | 3000–3200 Å  |
|--------------------|--------------|--------------|--------------|
| Tethys (leading)   | 0.52 ± 0.02  | 0.57 ± 0.02  | 0.61 ± 0.02  |
| Tethys (trailing)  | 0.39 ± 0.01  | 0.46 ± 0.01  | 0.50 ± 0.02  |
| Dione (leading)    | 0.58 ± 0.05  | 0.57 ± 0.03  | 0.60 ± 0.03  |
| Dione (trailing)   | 0.24 ± 0.05  | 0.32 ± 0.04  | 0.33 ± 0.04  |
| Rhea (leading)     | 0.43 ± 0.05  | 0.46 ± 0.03  | 0.46 ± 0.03  |
| Rhea (trailing)    | 0.25 ± 0.09  | 0.28 ± 0.04  | 0.30 ± 0.04  |
| Iapetus (leading)  | 0.040 ± 0.02 | 0.040 ± 0.01 | 0.041 ± 0.01 |
| Iapetus (trailing) | 0.298 ± 0.03 | 0.308 ± 0.02 | 0.32 ± 0.02  |

of other Saturnian satellites. The leading side albedo is  $\sim 6$  times lower than the trailing side albedo at the IUE wavelengths, somewhat greater than the 5 times darker reported at visual wavelengths. The spectral absorber that darkens the leading hemisphere of Iapetus is more absorbing toward shorter wavelengths. Efforts to identify this absorber should focus on a similar decrease in reflectance in the laboratory spectrum of any candidate absorber.

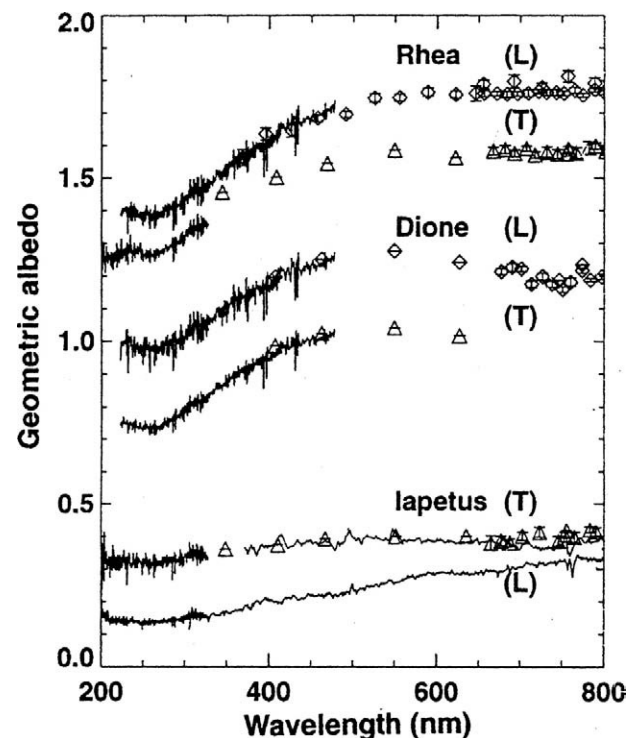
The broadband UV albedos reported by *IUE* and *HST* observations of the Saturnian satellites confirm the suggested differences in chemical/mineralogical composition on Dione, Rhea, and Iapetus that the longer wavelength observations imply. In the case of Dione, these observations indicate that there are no strong UV absorptions in the unidentified materials on the satellite's surface. In the case of Rhea and Iapetus, the UV absorption becomes greater toward longer wavelengths. This may be due to a gradual decrease in reflectance or may be the effect of an absorption band.

*HST* observations of Dione and Rhea (Figs. 12a and 12b) have detected an absorption similar to the 2600 Å absorption feature detected by *HST* on Ganymede and has been attributed to the presence of ozone on both satellites. Like Ganymede, Dione and Rhea orbit within the magnetosphere of their planet. Ozone on these satellites is likely a product of radiolysis. Iapetus, orbiting outside the magnetosphere, does not exhibit the 2600 Å absorption feature. However, because the hemispheric albedo dichotomy on Iapetus is greater at NUV wavelengths compared to the visual wavelengths, the possibility of an absorption feature somewhere between 2400 and 5600 Å can be inferred.

More recently, with the arrival of the *Cassini* spacecraft at the Saturn system, FUV measurements of the icy satellites have been made with the UVIS. The FUV spectra of the icy satellites are dominated by the strong water absorption feature at  $\sim 1650$  Å. At wavelengths shortward of  $\sim 1650$  Å, the icy satellites are extremely dark due to the presence of water ice, and in fact are darker than the surrounding interplanetary hydrogen (IPH). *Cassini* UVIS images of the icy satellites therefore reveal both the day and night sides of the icy satellites. Figure 13a displays the FUV reflectance spectrum of Phoebe, one of the outermost satellites of Saturn; Phoebe's spectrum is compared with the UVIS-measured reflectance spectrum of the Moon. Figure 13b shows the FUV image of Phoebe; the visible wavelength image is shown in Fig. 13c for comparison.

### 4.3 Enceladus

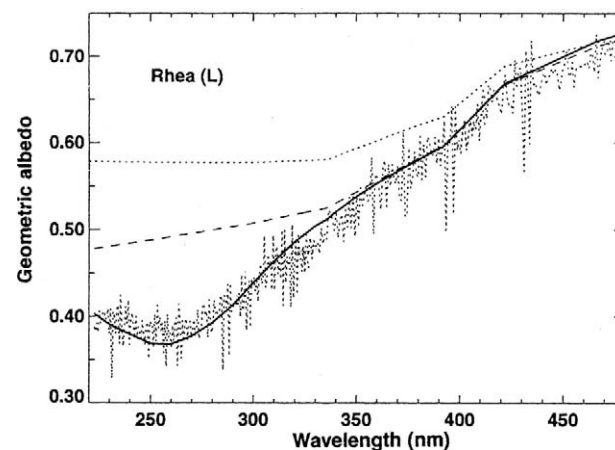
Enceladus, not easily observed from Earth orbit due to its proximity to Saturn and the rings, has recently been the target of key UV measurements from the *Cassini* spacecraft in orbit around Saturn. Enceladus has long intrigued scientists because it is the brightest object in the solar system; *Voyager* images revealed vast regions that were evidently crater-free, suggesting recent resurfacing by geologic



**FIGURE 12a** The NUV geometric albedos of the leading (L) and trailing (T) hemispheres of Rhea, Dione, and Iapetus as measured by *HST* FOS (thick solid lines). Also shown are longer wavelength albedos (thin lines and discrete points) from other sources. The Rhea albedos are offset by 1.0 and the Dione albedos are offset by 0.5. The spectrum of Iapetus' leading hemisphere has been scaled by a factor of 2.5. (Figure reproduced with permission from Nature Publishing Group.)

activity. Furthermore, its orbit at the densest part of the broad, tenuous E-ring has suggested that Enceladus could somehow be the source of the E-ring ice particles.

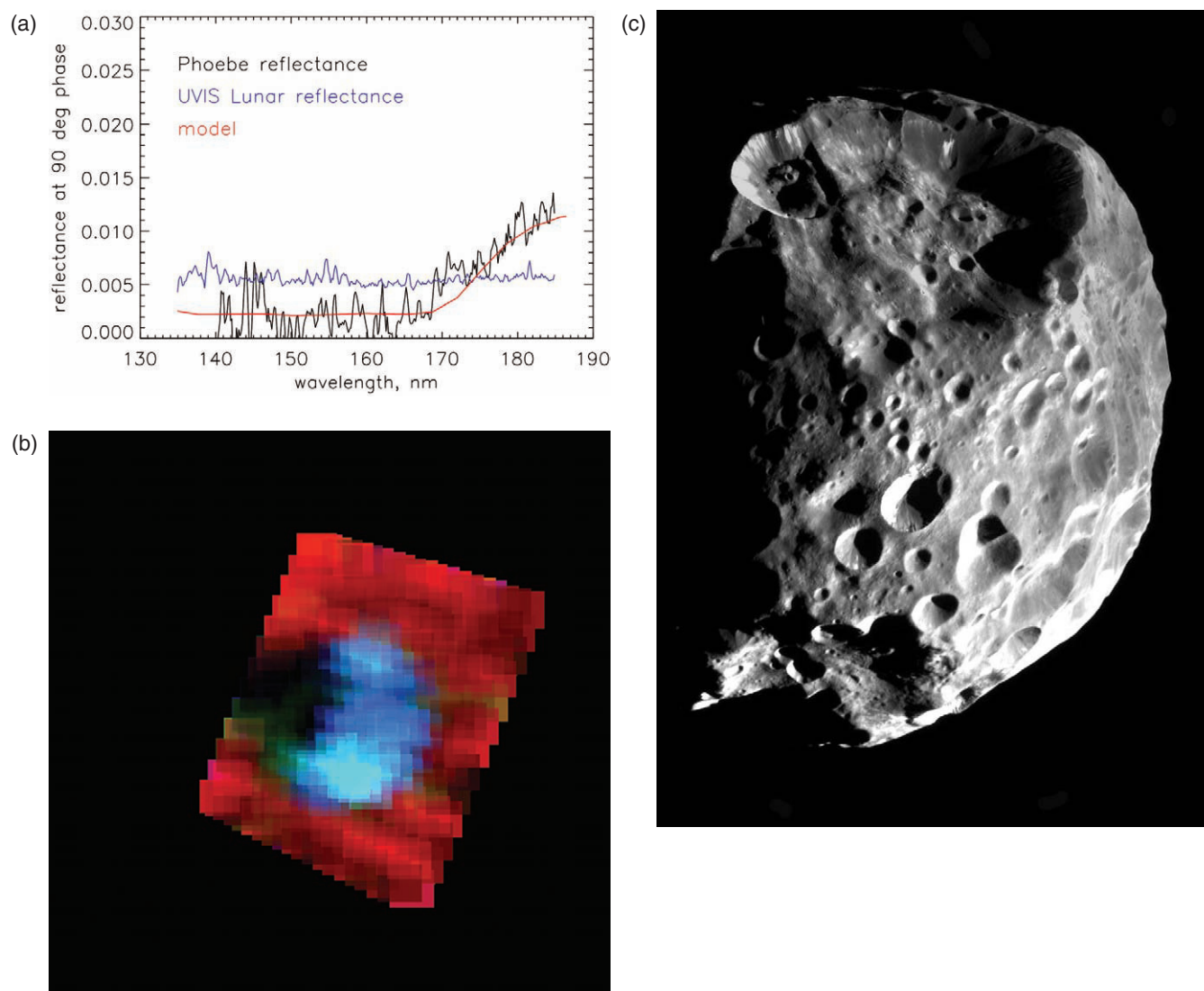
Ultraviolet measurements from *HST* detected the hydroxyl radical, OH, in emission (3085 Å) in the Saturn



**FIGURE 12b** Rhea's leading hemisphere albedo (from Fig. 12a) with a model including  $O_3$ . (Figure reproduced with permission from Nature Publishing Group.)



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**FIGURE 13** Saturn's moon Phoebe as measured by the *Cassini* UVIS. (a) Phoebe's FUV spectral reflectance measured at 90° **solar phase angle**. Phoebe's spectrum is overplotted with a model (red) that includes H<sub>2</sub>O frost and dark, carbonaceous material. Also shown is the lunar albedo (blue) as measured by UVIS at the same phase angle. Note that water ice on Phoebe makes it even darker than the Moon at short FUV wavelengths, and brighter at long FUV wavelengths. (b) FUV wavelength image of Phoebe. Red represents 1216 Å where interplanetary hydrogen is bright throughout the solar system. Phoebe's water ice makes it darker than the background IPH on both the illuminated and the dark hemispheres so that the entire disk is visible. Blue colors represent longer FUV wavelengths and show that the brightness varies across Phoebe's surface due to solar incidence, topography, and compositional variations. (c) Visible wavelength image of Phoebe from *Cassini*. (Figures reproduced with permission from AAAS/Science; ISS image courtesy of NASA.)

system, primarily near the orbit of Tethys. Similarly, *Cassini* UVIS measured neutral oxygen (at 1304 Å), in varying amounts, with the greatest abundances near the orbit of Enceladus. The presence of OH and O suggested that H<sub>2</sub>O is produced by erosion of the inner icy satellites of Saturn by micrometeoroid bombardment and is then broken down by photodissociation to produce the neutral species. However, the amounts of H<sub>2</sub>O necessary to produce the observed

OH and O abundances were not consistent with sputtering rates; an additional source of H<sub>2</sub>O was needed—and remained a mystery until *Cassini* observations of Enceladus in 2005.

The *Cassini* spacecraft, through unique multi-instrument observations, discovered active water plumes on Enceladus. A stellar occultation by *Cassini* UVIS measured the presence of water vapor above the limb of the

South Pole. Similar UV occultations of other regions of the moon had found no evidence of any gases, indicating that the vapor was locally confined to the south polar region. Surface temperatures, measured by the far-IR spectrometer, were found to be anomalously high; measurements by the magnetometer, mass spectrometer, dust detector, and near-IR spectrometer on *Cassini* confirmed the presence of gaseous species and ice grains being expelled from Enceladus' south polar hot spot. It is surprising that such a small, icy body is currently geologically active! The cause of the south polar hot spot and associated plumes is under investigation, and Enceladus remains a primary observational target of the *Cassini* mission.

#### 4.4 Uranian Satellites

The five major satellites of Uranus—Miranda, Ariel, Umbriel, Titania, and Oberon—are a suite of icy satellites that are situated at about the limit at which *IUE* was able to confidently return spectral information. They are so faint that it is not even possible to divide the *IUE* wavelength range into several bands, as was done with the jovian and Saturnian satellites. All the spectral information is integrated into one wavelength range, and a geometric albedo can be determined.

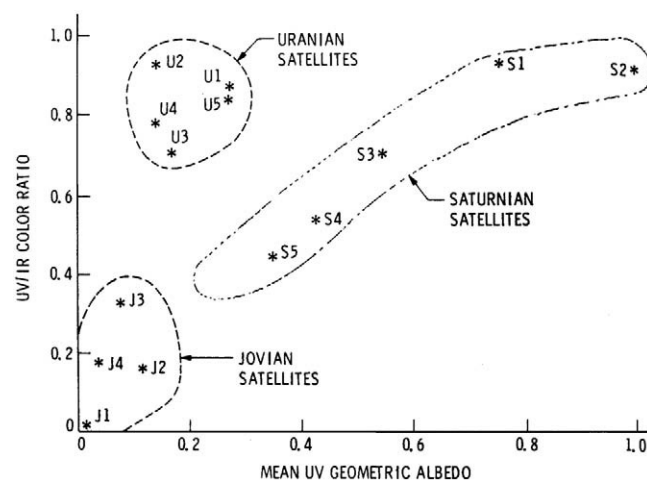
The Uranian satellites are in an orbital plane that is parallel with the Uranian equator, and the pole of Uranus' orbit is tilted such that, at the present time, it is pointed toward Earth. Therefore, only the poles of one hemisphere of the satellites of Uranus are observable with *IUE*, and hence it is not possible to construct orbital phase curves and leading/trailing side ratio spectra.

*IUE* was able to observe Oberon, Uranus's brightest satellite. The *IUE* result proved to be an important and independent confirmation of results from the *Voyager 2* photopolarimeter experiment. The ultraviolet geometric albedo of Oberon was found to be 0.19–0.025, an excellent confirmation of the earlier *Voyager 2* PPS result of 0.17.

Spectra from 2200 to 4800 Å were obtained with the *HST* FOC for the Uranian satellites Ariel, Titania, and Oberon. The inner Uranian satellites Miranda and Puck were also observed from 2500 to 8000 Å with the *HST* FOC. The geometric albedos for Ariel, Titania, and Oberon display a broad, weak absorption at 2800 Å, similar to the feature seen on Europa and Callisto. Although this absorption feature on the Galilean satellites has been attributed to SO<sub>2</sub>, it has been attributed to OH on the Uranian satellites. Both SO<sub>2</sub> and OH produce an absorption feature near 2800 Å, however, the molecule OH (a by-product of the photolysis and radiolysis of water) is unstable at the surface temperatures of the Galilean satellites but is stable at the colder surface temperatures of the Uranian satellites. No detection of the 2600 Å ozone feature seen on Ganymede, Dione, and Rhea has been detected in any of the Uranian satellite spectra.

#### 4.5 A Comparison of Icy Satellite Systems

The ultraviolet observations of planetary satellites can be integrated with the results of observations at longer wavelengths to provide a comparative assessment of the families of large planetary satellites in the solar system. The *IUE*- and *HST*-determined photometric properties of the larger planetary satellites of Jupiter, Saturn, and Uranus are shown in Fig. 14 as a plot of ultraviolet-to-infrared color ratio versus ultraviolet geometric albedo. The geometric albedos of the Saturnian satellites indicate that in this system there is a wide variation in UV geometric albedo. The Galilean and Uranian satellites have photometric properties that are common within each group. This is consistent with the hypothesis that the surface modification processes that have occurred are similar within the Galilean and Uranian satellite systems, but the two systems have surface modification processes that are distinct from each other. The diverse nature of the photometric properties of the Saturnian satellites



**FIGURE 14** Comparisons of geometric albedos of the Galilean, Saturnian, and Uranian satellites. For the Galilean satellites and for Dione, Rhea, and Tethys, the UV albedos are from *IUE* and *HST*. For Mimas and Enceladus, the UV albedos are from *Voyager* images and are of longer wavelength (3500 Å) than those from *IUE* and *HST*. The infrared albedos of the Galilean and Saturnian satellites are from ground-based spectrophotometry. All the Uranian satellite data are from the *Voyager* Photopolarimeter. With the exception of the slight difference in wavelength from Mimas and Enceladus noted previously, all the wavelength ranges are similar. Jovian and Uranian satellites each have distinct color ratios that distinguish the two groups of satellites from each other. The Saturnian satellites have little albedo similarity among themselves. This is consistent with the hypothesis that the surfaces of all the icy Galilean satellites are being modified by a common process. Likewise, the Uranian satellite surfaces may also have a common process of surface modification that is different from the jovian system. The Saturnian satellite system may not have a common process of surface modification, or the system may have been disturbed and the albedos of the satellites altered.

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suggests that no common surface modification process is altering the surfaces of the satellites or, alternatively, that the entire satellite system was recently modified in some way and the slow process of space weathering has not had time to restore the surfaces to a common photometric state. A possibility is that coating by E-ring grains, an ongoing process, effectively keeps the optical surface of the Saturnian satellites young and unweathered.

In general, the NUV spectra of icy satellites are dominated by weathering products. Radiolysis and photolysis are extremely important processes at the surfaces of these satellites, and products of these processes are apparent at NUV wavelengths. This is evidenced by the presence of  $\text{SO}_2$ ,  $\text{O}_3$ , and  $\text{H}_2\text{O}_2$  in the surfaces of the icy Galilean satellites (and  $\text{O}_3$  in the surfaces of some of the icy Saturnian satellites). The icy Galilean satellites are all relatively dark at UV wavelengths. However, water ice, the primary constituent of these surfaces, is bright in the NUV. Therefore, another material must be responsible for the ultraviolet absorption of the icy Galilean satellites. The most likely darkening agents are elemental sulfur and sulfur-bearing compounds originating from the very young and active surface of Io, which are transported as ions outward from Io's orbit by jovian magnetospheric processes. These energetic ions and neutrals interact with the icy surfaces of Europa, Ganymede, and Callisto and cause the ices to become darkened at UV wavelengths. This process competes with other processes of surface modification such as infall of interplanetary debris.

In contrast to Jupiter's UV-dark icy satellites, Saturn's icy satellites, particularly those orbiting closer to Saturn, are relatively bright. This may be related to the presence of the large and tenuous E-ring. Mimas, Enceladus, Tethys, Dione, and possibly Rhea all orbit Saturn within this broad ring of tiny icy grains. The relative velocities between the E-ring particles and the icy satellites may explain the overall brightness, as well as the large-scale longitudinal albedo patterns on the icy satellites. Mimas and Enceladus are both slightly darker on the leading hemispheres than on the trailing hemispheres, possibly because the E-ring particles sweep by the trailing hemispheres, brightening them. The leading hemispheres of Tethys and Dione are brighter than the trailing hemispheres because their leading hemispheres sweep by the E-ring particles. Furthermore, Saturn's magnetosphere is different from Jupiter's and appears to be dominated by neutrals rather than by electrons and other charged particles. This difference may have an effect on the ice chemistry that occurs within the surfaces of the icy satellites because much of the ice chemistry occurring on the icy Galilean satellites is the result of charged particle bombardment. The Saturnian satellites are less spectrally red in the NUV than the Galilean satellites, and  $\text{H}_2\text{O}_2$  does not appear to be present, in contrast to the Jovian satellites. These differences could be a result of the different charged particle environments in the Saturn and Jupiter magnetospheres.

### 4.6 Pluto and Charon

The first spatially resolvable images of Pluto and its satellite Charon were obtained by the *HST* FOC at visible (4100 Å) and ultraviolet (2780 Å) band passes. The image resolution is sufficient to show the presence of large, longitudinally asymmetric polar cap regions in addition to a variety of albedo markings. The combination of UV and visible images were used to look for regions of clean ice and nonclean (contaminated) ice—either radiation darkened or sites where atmospheric chemistry products were deposited. No positive identification of solids on the surfaces of Pluto and Charon has been made at UV wavelengths. The cleanest ice (bright in UV and VIS bandpasses) was found in a location at the equator, though overall the equator was found to be heavier in contaminated ice than mid latitudes. The north polar region was found to have the cleanest ice.

### 4.7 Asteroids and Comets

At ultraviolet wavelengths, asteroids have been studied by *IUE* and *HST* from Earth-based orbit. The *Galileo* spacecraft obtained NUV spectra of Ida and Gaspra during its travels through the Asteroid Belt. Finally, the *Mariner 9* UVS obtained spectra of Phobos and Deimos, the martian moons that are likely captured asteroids.

The *IUE* satellite obtained ultraviolet observations of ~45 main-belt asteroids in the wavelength range between 2300 and 3250 Å. The geometric albedos for these objects are consistently low, and three major asteroid taxonomic classes seen in the visible persist into the ultraviolet. Analysis of the *IUE* asteroid data shows that the asteroids observed have ultraviolet albedos that range from 0.02 for C-class asteroids to 0.08 for M-class asteroids; albedos of S-class asteroids are intermediate. Analysis of a set of *IUE* Vesta observations covering more than one rotation of Vesta indicates that this unusual asteroid displays UV albedo variations across the surface such that the UV lightcurve of Vesta is opposite that of the visible lightcurve. Such a spectral reversal is consistent with a hemispheric dichotomy in composition and/or a variation in geologic age (due to space weathering) across the surface. A study of *IUE* measurements of S-class asteroids has found that the UV spectral slope may be an important indicator of space weathering and ultimately exposure age; the strong decrease in albedo that is typical in silicates at NUV wavelengths appears to lessen with exposure. Measurements at visible-infrared wavelengths suggested the presence of hydrated minerals on Ceres; a search for OH (emissions at ~3085 Å) in *IUE* spectra, however, found none. *HST* FOC, and more recently ACS, images of Ceres at UV wavelengths were the first well-resolved images of this largest asteroid. Albedo variations are detectable across the surface. The brightness of the surface in the FUV may rule out the presence of a large amount of water ice; an absorption band



with a central wavelength of 2800 Å may be present but its cause is unidentified. Analysis of *Mariner 9* UVS spectra of Phobos and Deimos show these bodies to be spectrally similar to carbonaceous chondrites. However, analysis of *HST* FOS data of these moons at UV-visible wavelengths, compared with FOS spectra of a C-type asteroid and a D-type asteroids, showed the martian moons to be more similar to the D-type asteroid than to the C-type asteroid.

Both *HST* and *IUE* observed the Centaur asteroid 2060 Chiron, a possible former resident of the Kuiper Belt. Neither instrument detected emission from gaseous species at ultraviolet wavelengths, in contrast to CN emissions that have been reported at visible wavelengths. The UV albedo of Chiron is similar to that of some of the Saturnian and Uranian satellites. In particular, Chiron's UV/IR color and ultraviolet albedo are very similar to those of Dione. [See KUIPER BELT OBJECTS: PHYSICAL STUDIES.]

Observations of comets at ultraviolet wavelengths are extremely useful for measuring fluorescence of solar photons by important atomic and molecular species, and thereby studying relative abundances of the vaporizing species and probing the photochemical and physical processes acting in the densest regions of the coma. UV observations of comets were first accomplished by sounding rockets and the OAO satellite prior to the launch of *IUE*. These observations established the emission of hydroxyl ion at near the limit of ground-based observations, 3085 Å. This is consistent with a cometary composition dominated by water ice; the hydroxyl ion is a product of exposure to solar radiation.

*IUE* observed more than 50 comets (~400 individual spectra). *IUE*'s photometric constancy provided the ability to compare observations of comets that appeared several years apart. Those observed range from short-period comets with aphelion near Jupiter to long-period comets that may be first-time visitors to our solar system.

All the comets observed by *IUE* have displayed the 3085 Å hydroxyl line, which is consistent with water ice being a major part of comet composition. Although all comets appear to have similar principal compositional components (water), each has different amounts of trace components, including carbon dioxide, ammonia, and methane, detected by *IUE* and *HST*. Gas production rates have been derived for species such as H<sub>2</sub>O, CS<sub>2</sub>, and NH<sub>3</sub>. Several comets that were observed over a long period of time exhibited differences in their dust-to-gas ratios from one observation to the next, consistent with a variation as a function of heliocentric distance.

The first detection of diatomic elemental sulfur in a comet was seen in comet IRAS-Araki-Alcock. The lifetime of the diatomic sulfur in the cometary atmosphere is quite short (~500 seconds). This makes sulfur a useful tracer of the dynamics of the tenuous cometary atmosphere, which appears during the short time that the comet is near the Sun. Analysis of the S I triplet emission band near 1814 Å in cometary comae spectra taken with *IUE* and the *HST* FOS

shows that cometary sulfur, which is present and stored in a variety of volatile species, is depleted in abundance compared to solar abundances. The detection of CS at ultraviolet wavelengths in comae is attributed to the presence of CS<sub>2</sub> in the comet. Sulfur detected in the comae in excess of the sulfur attributable to CS<sub>2</sub> is assumed to originate from H<sub>2</sub>S and nuclear atomic sulfur in the comet. Using this assumption, models have been used to measure total sulfur versus water abundances, which range between ~0.001 and ~0.01. [See PHYSICS AND CHEMISTRY OF COMETS.]

Ultraviolet observations using *IUE*, *HST*, and *FUSE* have also detected ultraviolet CO Cameron band emissions from comets, which is useful for measuring the CO<sub>2</sub> production rate. This rate derived from *IUE* observations of comet 1P/Halley agrees with the rate measured in situ by the spacecraft *Giotto*. These *HST* and *IUE* observations suggest that the level of activity of a comet may be linked to its CO abundance; however, this is based on a small sample of the comet population. *FUSE* measurements of C/2001 A2 (LINEAR) displayed H<sub>2</sub> emission lines of the Lyman system at 1071.6, 1118.6, and 1166.8 Å, in addition to CO features that suggested both a hot and a cold component of CO, the hot component likely being due to excitation of CO<sub>2</sub>, the cold component being attributed to fluorescent scattering of CO or to electron impact excitation of CO.

#### 4.8 The Moon and Mercury

The first UV observations of the Moon were made at FUV wavelengths using the instrument aboard the *Apollo 17* orbiter. It was noted in these measurements that the lunar maria regions, darker than the highlands at visible wavelengths, are brighter than the highlands in the FUV. This was the first indication of the so-called spectral reversal, which was also detected at EUV wavelengths using measurements by the *EUVE*. This phenomenon is attributed to the concept that FUV measurements probe just the outer layers of the grains (surface scattering, as opposed to volume scattering measured at longer wavelengths), and that space weathering processes may cause the lunar grains to be covered with a fine coating. Lunar samples measured in the laboratory support this idea: Lunar soils (presumably more weathered) show the spectral reversal, while ground-up lunar rocks (presumably less weathered) do not. *Galileo* UVS measurements in the NUV showed that the maria are darker than the highlands and that the spectral reversal must occur at a wavelength shorter than ~2200 Å. The *HUT* measurement of the lunar surface (a region near Flammarion-C, a border area between mare and highlands) at FUV wavelengths indicated an albedo of ~4% with a slight increase in brightness toward shorter wavelengths. Because of the different spectral behavior at UV versus visible wavelengths, ratio images of UV to visible color images and visible reflectance spectra are used to map spectral trends related to opaque mineral abundance and the combined effects of

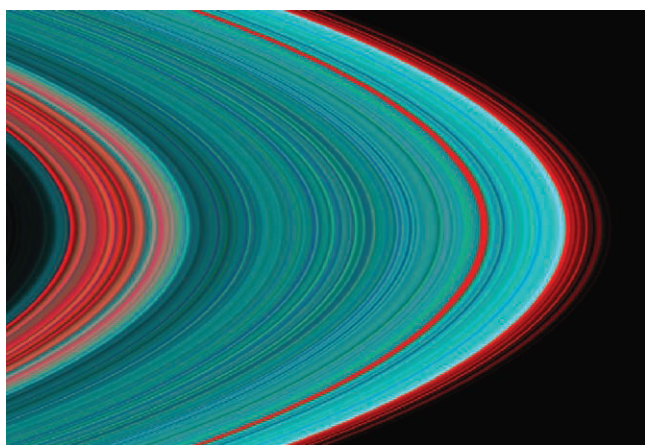
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FeO content and soil maturity. From the *Apollo* samples, it is known that the dominant opaque mineral is ilmenite, which is high in Ti content. Thus, UV/visible ratio images have been used to map Ti content variation in the lunar mare basalts.

The *Mariner 10* spacecraft carried a color imager that included a near UV filter (3550 Å). Mercury image ratios (UV/visible) have been used to map spectral trends associated with geologic features, using similar methods as used on lunar images. A lower UV/visible ratio suggests more FeO, or more mature soil. Spectrally neutral opaque minerals (such as ilmenite) tend to lead to a higher UV/visible ratio. Mercurian regions believed to be volcanic in origin have been found to have FeO amounts slightly less than average, consistent with ancient lava flows.

### 4.9 Planetary Rings

The rings of Saturn were successfully observed by *IUE* in a series of observations between 1982 and 1985. The spectrum of the rings in the 1600–3100 Å range is dominated by the water ice absorption edge at ~1650 Å. More recently, *Cassini* UVIS has made higher resolution observations of Saturn's rings; an image is shown in Fig. 15. This image shows a combination of the UV reflectance and transmission of the ring system. The red-colored region at the left is



**FIGURE 15** Saturn's rings as imaged by *Cassini* UVIS. This false color two-dimensional representation of Saturn's Cassini Division and A ring was generated from UVIS data obtained during a radial scan of the rings immediately after Saturn Orbit Insertion as *Cassini* flew over the rings. To generate the image, azimuthal symmetry was assumed. Although there are azimuthal variations in the structure of the rings, they are smaller than the 100 km resolution of this image. Red represents Lyman-alpha emission from interplanetary hydrogen (1216 Å) and shines through gaps and optically thin parts of the ring. Green and blue represent reflection of solar ultraviolet light longward of the water ice absorption edge near 1650 Å. (Figure courtesy J. Colwell.)

the Cassini Division with a mean opacity of about 0.1, and the thin bright band near the outer edge of the rings is the 300 km wide Encke gap. Brighter blue-green regions indicate cleaner water ice (less absorption by non-ice species). The A ring material is cleaner than the *Cassini* Division and the abundance of water ice is seen to increase near the outer edge of the A ring.

## 5. Conclusions

The importance of ultraviolet solar system science has been exhibited through discoveries and continuing studies spanning the topics of atmospheric and auroral science, surface composition and space weathering. Ultraviolet observations of solar system surfaces and atmospheres have been made possible by the *IUE* and *HST* orbiting telescopes, along with *FUSE*, *HUT*, and *EUVE*, and have been substantially complemented by interplanetary missions such as *Voyager*, *Mariner*, *Galileo*, and *Cassini*. The *IUE* spacecraft provided the astronomical community with the first stable long-term (spanning nearly two decades) observing platform in space, from which astronomers have been able to study regions of the spectrum that are inaccessible from telescopes on Earth's surface. This foundation, with the support of ultraviolet spectrometers incorporated into the payloads of deep space missions, filled an observational void that had existed since the dawn of astronomy. These observations have led to important new discoveries and have provided tests of physical models that have been developed based on ground-based observations. *IUE*'s observing capability was surpassed by *HST*, which has provided the astronomical community with the opportunity to look at fainter and more distant solar system objects and has led to new discoveries in the ultraviolet spectrum. The future of Earth-based orbiting UV telescopes is unclear, but such UV instruments, with ever-improving spectroscopic and imaging capabilities, are vital to understanding solar system objects and complementing longer-wavelength observations.

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